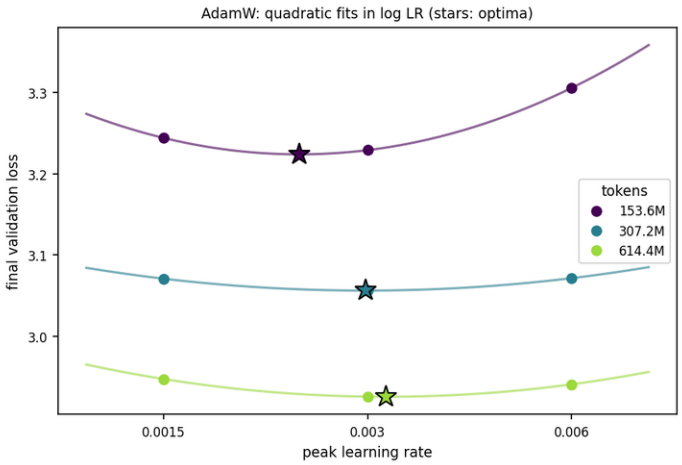
**Assignment 2: Hyperparameter Scaling** (Kesavan Ramakrishnan)

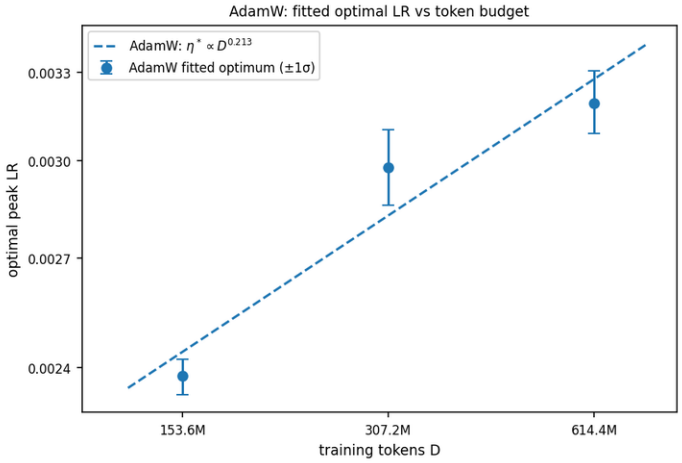
## Problem 1: When can training on less data guide large scale runs?

### (a) Small budgets




Final validation loss, supplied AdamW runs (WD 0.1; lowest per budget in bold):

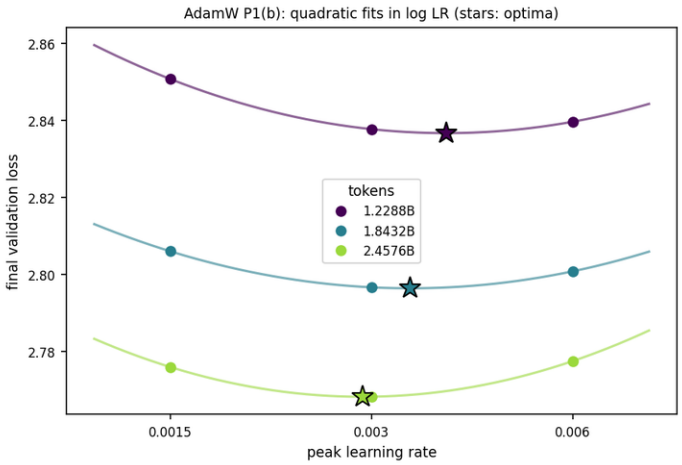
| Peak LR | 153.6M | 307.2M | 614.4M |
|---|---|---|---|
| 0.0015 | 3.2442 | 3.0709 | 2.9473 |
| 0.003 | **3.2291** | **3.0562** | **2.9256** |
| 0.006 | 3.3056 | 3.0715 | 2.9408 |



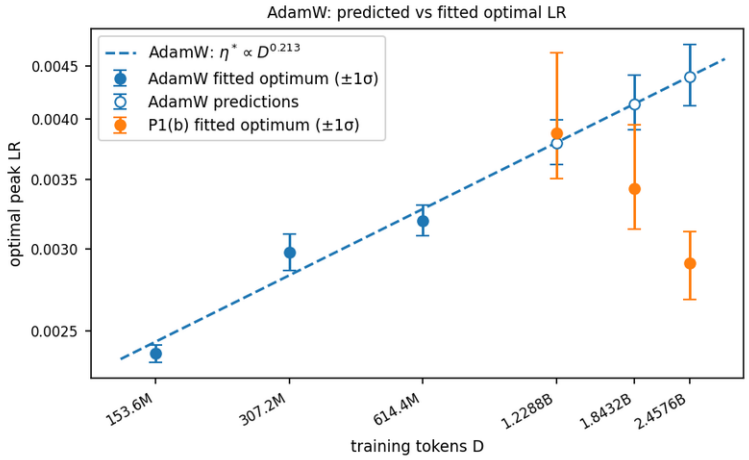

Proposed rule (power law through the fitted optima): η*(D) = 0.00328 · (D / 614.4M)^0.213. The optimal LR rises with token count.

### (b) Predictions at larger budgets

Recorded before inspecting the data, from the rule in (a): 0.00380 at 1.2288B, 0.00414 at 1.8432B, 0.00440 at 2.4576B.

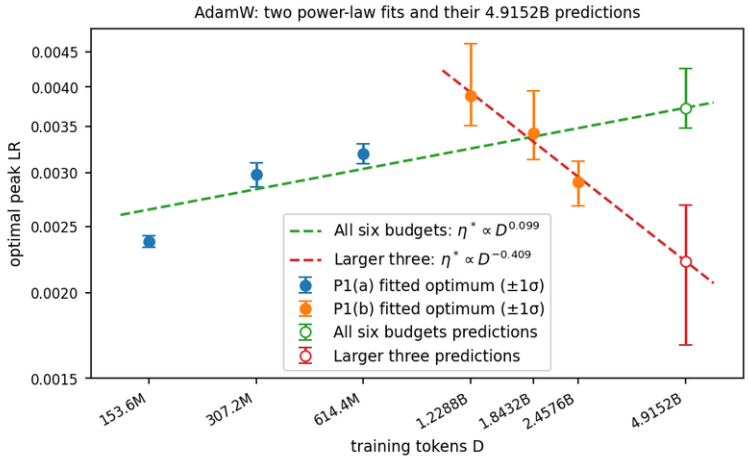


| Peak LR | 1.2288B | 1.8432B | 2.4576B |
|---|---|---|---|
| 0.0015 | 2.8508 | 2.8060 | 2.7760 |
| 0.003 | **2.8378** | **2.7967** | **2.7683** |
| 0.006 | 2.8397 | 2.8008 | 2.7775 |



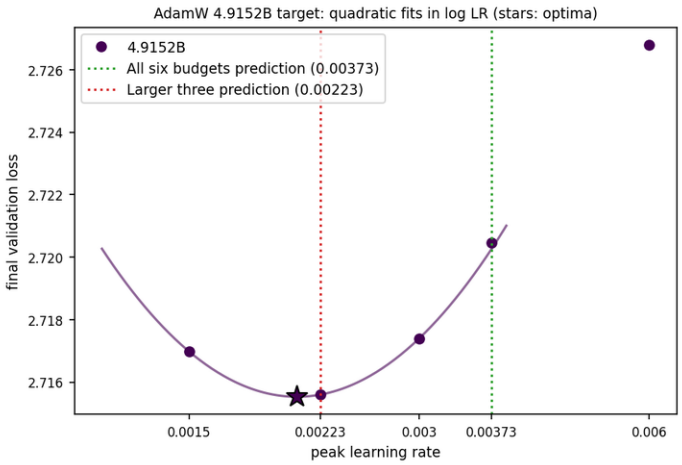

The trend does not continue. The prediction holds at 1.2288B (0.00380 vs fitted 0.00388), but the fitted optimum then falls to 0.00343 and 0.00291, so the rule overpredicts by 21% and 51%.

**(c) Two scaling rules**




| Fit | Rule | Predicted η* at 4.9152B |
|---|---|---|
| All six budgets | η* ∝ D^0.099 | 0.00373 |
| Larger three (1.2288B–2.4576B) | η* ∝ D^−0.409 | 0.00223 |

The larger-three fit should do better: the optimum stopped rising near 1.2B tokens, and the all-six fit averages rising and falling optima, so it still predicts an increase.

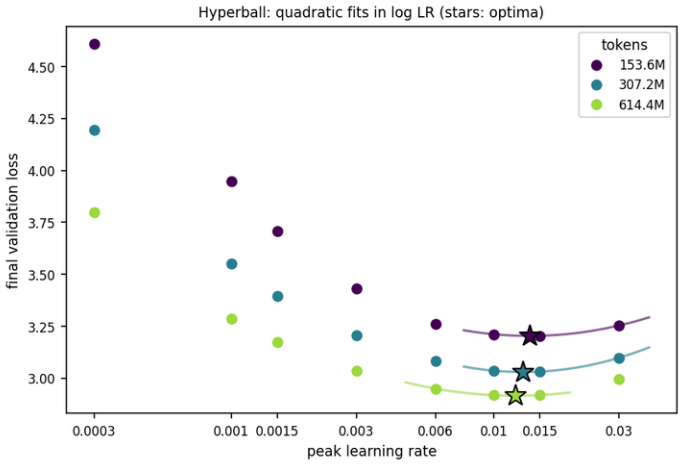


| Peak LR | Source | Final val loss | Gap vs best grid LR |
|---|---|---|---|
| 0.0015 | grid | 2.7170 | best grid |
| 0.00223 | larger-three | **2.7156** | −0.0014 |
| 0.003 | grid | 2.7174 | +0.0004 |
| 0.00373 | all-six | 2.7205 | +0.0035 |
| 0.006 | grid | 2.7268 | +0.0098 |

The fitted target optimum is 0.00208 (±1σ 0.00182–0.00235); the larger-three prediction is 7% above it, the all-six prediction 79% above.

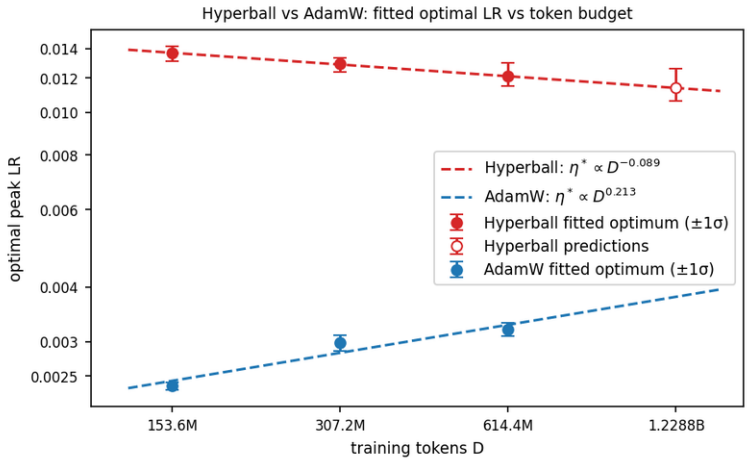
- **Gains from the smaller-budget fits:** none. The larger-three gain (0.0014) is within seed noise (SD 0.0024), and the all-six fit, which leans on the smallest budgets, was 0.0035 worse.
- **Bias–variance:** the all-six fit has low variance (exponent ±1σ 0.08–0.14) but is biased, since one power law cannot follow an optimum that rises then falls. The larger-three fit has less bias but high variance (−0.71 to −0.24). Bias dominated.
- **Extrapolation:** the local fit held for one 2× step. Because the trend reversed inside the source range, trust a fit only a short distance past its data. The loss is flat near the optimum (LRs 0.0015–0.003 within 0.0018 of the best), so modest LR errors cost little here.

**(d) Hyperball versus AdamW**




Supplied Hyperball runs (WD 0; lowest per budget in bold):

| Peak LR | 153.6M | 307.2M | 614.4M |
|---|---|---|---|
| 0.0003 | 4.6079 | 4.1946 | 3.8001 |
| 0.001 | 3.9460 | 3.5495 | 3.2863 |
| 0.0015 | 3.7064 | 3.3939 | 3.1730 |
| 0.003 | 3.4326 | 3.2063 | 3.0364 |
| 0.006 | 3.2596 | 3.0835 | 2.9494 |
| 0.01 | 3.2113 | 3.0357 | **2.9176** |
| 0.015 | **3.2040** | **3.0315** | 2.9183 |
| 0.03 | 3.2533 | 3.0981 | 2.9962 |

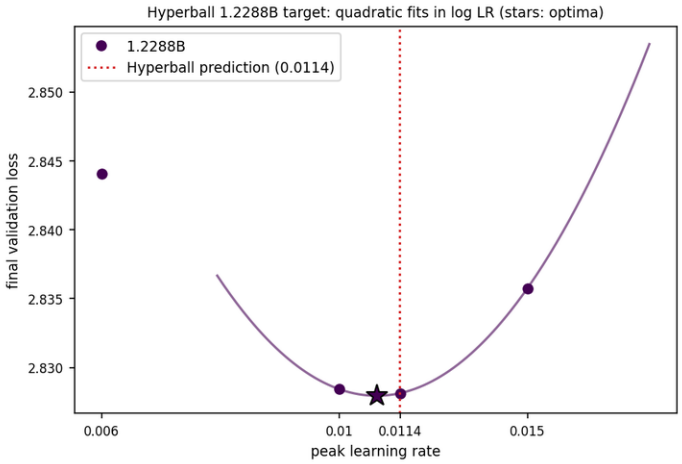



| Tokens | Hyperball η* | AdamW η* |
|---|---|---|
| 153.6M | 0.0137 | 0.00238 |
| 307.2M | 0.0129 | 0.00298 |
| 614.4M | 0.0121 | 0.00319 |
| Rule | ∝ D^−0.089 | ∝ D^0.213 |

Hyperball's optimum falls slowly (×0.94 per doubling) while AdamW's rises (×1.16). Hyperball's two doublings give nearly the same ratio (0.944,

0.937); AdamW's do not (1.25, 1.07).

Predicted optimum at 1.2288B, recorded before running the target: 0.0114.




| Peak LR | Final val loss | Gap vs best sampled |
|---|---|---|
| 0.006 | 2.8441 | +0.0160 |
| 0.01 | 2.8284 | +0.0003 |
| 0.0114 (predicted) | **2.8281** | best |
| 0.015 | 2.8358 | +0.0076 |

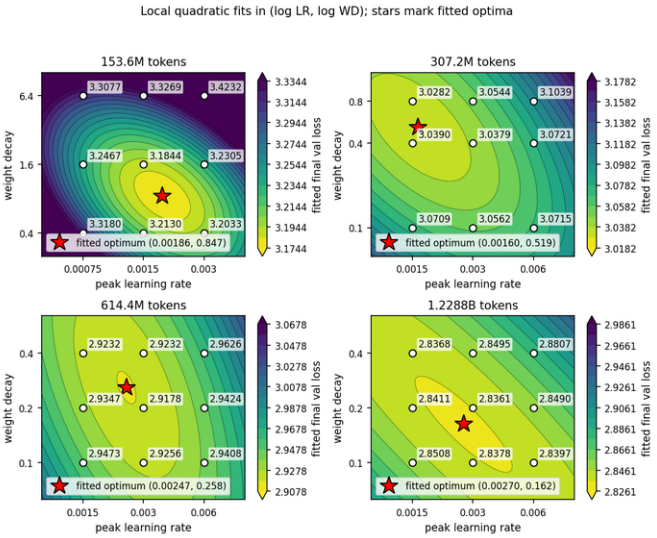
- **Prediction vs target:** the predicted LR gave the best sampled loss; the fitted optimum is 0.0108 ($\pm 1\sigma$ 0.0092–0.0118), 5% lower, and LR 0.01 is within noise. Hyperball's best loss is 0.0097 below AdamW's best sampled loss at this budget (2.8378).
- **More regular:** yes. Per doubling the Hyperball optimum changes by 0.94, 0.94, 0.90; AdamW's changed by 1.25, 1.07, 1.22, then reversed (0.75). But AdamW's rule also held at 1.2288B, so one target cannot show whether Hyperball keeps its trend past where AdamW's broke.
- **Uncertainty:** three budgets give an exponent of $-0.089$ ($\pm 1\sigma$ $-0.13$ to $-0.02$), so the direction is clear and the rate is not. Fitted optima move ±4–6% under seed noise, and four points give $-0.134$. AdamW shows a trend can reverse just past the source range.

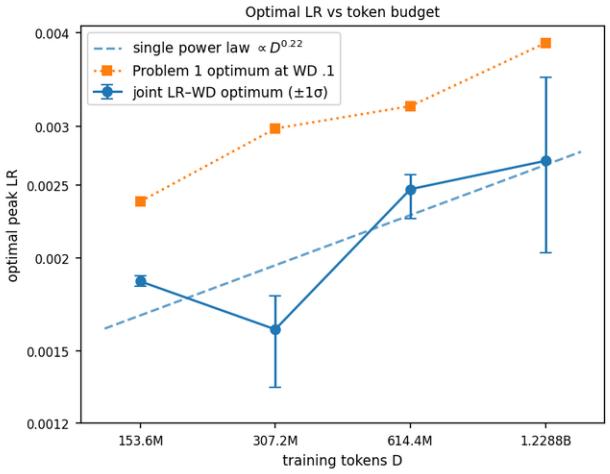
**Synthesis**

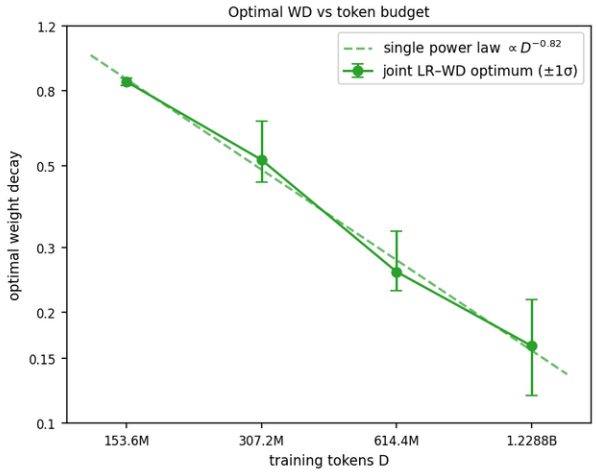
AdamW's optimal LR rose with budget up to about 1.2B tokens (0.00238 at 153.6M to 0.00388 at 1.2288B), then fell (0.00291 at 2.4576B, 0.00208 at 4.9152B), so larger budgets preferred lower LRs. Hyperball's fell slowly and steadily over its whole range (0.0137 to 0.0108, about ×0.94 per doubling). Because the AdamW trend reversed, extrapolating from small budgets failed: the rule from (a) overpredicted the 2.4576B optimum by 51%, and the all-six fit predicted an LR 79% above the 4.9152B optimum, while the fit on the three largest budgets came within 7%. Far-out predictions need source points near the target budget, and a power law should be trusted only a short distance past its data. Hyperball looked more regular, but it passed only one 2× extrapolation, as AdamW did before its trend broke, so three small budgets cannot rule out a later bend.

## Problem 2: How should learning rate and weight decay scale together?

### (a) LR and WD vary together

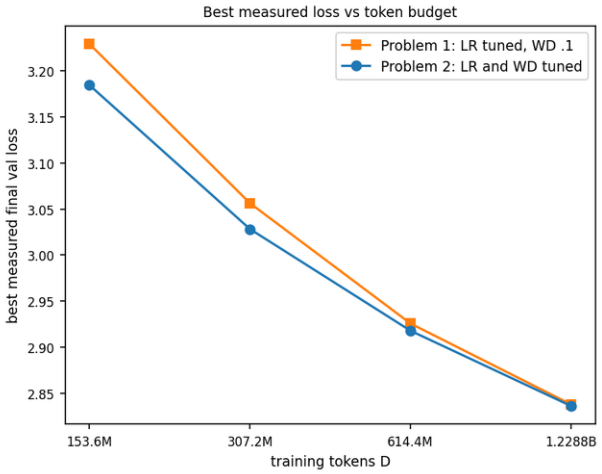


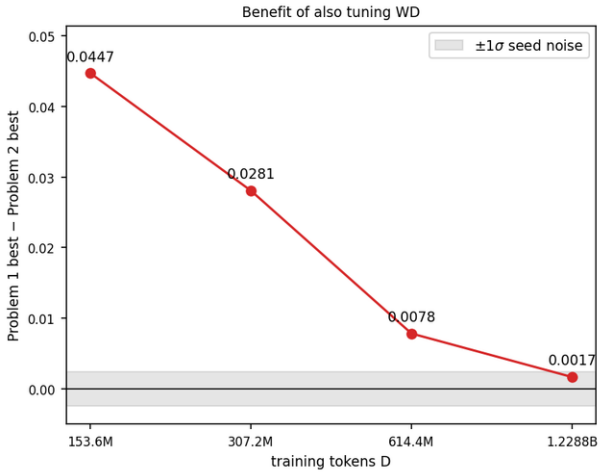



| Tokens | Fitted η* (±1σ) | Fitted λ* (±1σ) | Problem 1 η* (WD 0.1) |
|---|---|---|---|
| 153.6M | 0.00186 (0.00183–0.00189) | 0.847 (0.825–0.867) | 0.00238 |
| 307.2M | 0.00160 (0.00134–0.00178) | 0.519 (0.452–0.660) | 0.00298 |
| 614.4M | 0.00247 (0.00226–0.00258) | 0.258 (0.229–0.332) | 0.00319 |
| 1.2288B | 0.00270 (0.00203–0.00349) | 0.162 (0.119–0.216) | 0.00388 |
| Single power law | ∝ D^0.22 | ∝ D^−0.82 | |

- **Trends:** optimal WD falls steadily (0.85 to 0.16, ×0.50–0.63 per doubling). Optimal LR rises overall (0.00186 to 0.00270) but not monotonically: it dips to 0.00160 at 307.2M.
- **Compared with Problem 1:** the LR rises at a similar rate (D^0.22 vs D^0.21), but the joint optimum is 22–46% lower at every budget, where WD is also above 0.1.
- **Single power law:** WD yes (steady factor per doubling, close to D^−0.82). LR no: the 307.2M dip breaks the trend and the 1.2288B optimum is poorly constrained (±1σ 0.0020–0.0035).

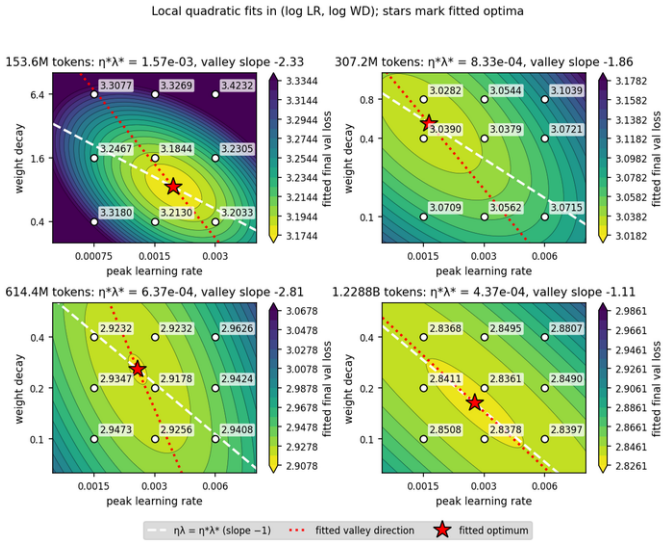



| Tokens | Best joint (LR, WD) | Best joint loss | Best Problem 1 loss (LR 0.003, WD 0.1) | Benefit of tuning WD |
|---|---|---|---|---|
| 153.6M | (0.0015, 1.6) | 3.1844 | 3.2291 | 0.0447 |
| 307.2M | (0.0015, 0.8) | 3.0282 | 3.0562 | 0.0281 |
| 614.4M | (0.003, 0.2) | 2.9178 | 2.9256 | 0.0078 |
| 1.2288B | (0.003, 0.2) | 2.8361 | 2.8378 | 0.0017 |

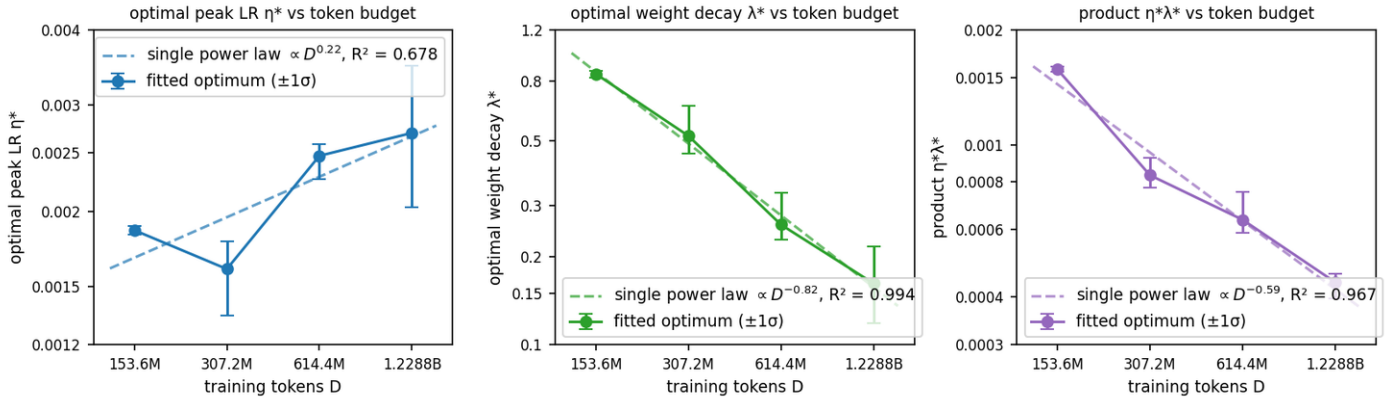
The benefit of tuning WD shrinks with budget, from 0.0447 to 0.0017 (within seed noise, SD 0.0024), because the optimal WD falls toward the 0.1 used in Problem 1 (0.16 at 1.2288B).

### (b) Joint scaling law




Fitted loss rise for a 2× move away from the optimum. The valley slope is d log λ / d log η along the fitted surface's flattest direction; the line ηλ = const has slope −1.

| Tokens | Valley slope | Along ηλ = η*λ* | LR only | WD only |
|---|---|---|---|---|
| 153.6M | −2.33 | +0.0376 | +0.0468 | +0.0195 |
| 307.2M | −1.86 | +0.0098 | +0.0148 | +0.0069 |
| 614.4M | −2.81 | +0.0136 | +0.0196 | +0.0055 |
| 1.2288B | −1.11 | +0.0020 | +0.0086 | +0.0072 |




| Variable | Exponent | R² (±1σ) | Change per doubling |
|---|---|---|---|
| η* | +0.22 | 0.678 (0.25–0.78) | ×0.86, ×1.54, ×1.09 |
| λ* | −0.82 | 0.994 (0.82–0.99) | ×0.61, ×0.50, ×0.63 |
| η*λ* | −0.59 | 0.967 (0.90–0.98) | ×0.53, ×0.77, ×0.69 |

- **Compensation:** partly. Moving along ηλ = η*λ* costs less than changing LR alone at every budget (e.g. +0.0020 vs +0.0086 at 1.2288B).
- **Product line vs contours:** it matches the valley only at 1.2288B (slope −1.11). At 153.6M–614.4M the valley is steeper (−1.9 to −2.8), so WD must move more than 1/LR to compensate, and a WD-only move (+0.0195, +0.0069, +0.0055) is cheaper than moving along the product line.
- **Regularity:** the product is more regular than η* (×0.53–0.77 vs ×0.86–1.54 per doubling) but not more than λ* (×0.50–0.63).
- **Tightest power law:** λ* ($R^2$ 0.994), then η*λ* (0.967), then η* (0.678). Under seed noise the product is the more stable of the top two (0.90–0.98 vs 0.82–0.99), but with four points the gap is not significant.

### (c) Test the joint rule

**Prediction.** The product law fitted on 153.6M–1.2288B is η*λ* ∝ D^−0.593, giving η*λ* = 2.782e−4 at 2.4576B; at LR 0.003 this is WD = 2.782e−4 / 0.003 = 0.0927.

Final validation loss at 2.4576B (gap = baseline minus predicted; positive means the prediction is better):

| Run | LR | WD | Source | Final val loss | Gap |
|---|---|---|---|---|---|
| Predicted (product law) | 0.003 | 0.0927 | new | 2.7667 | |
| Baseline (i): best 153.6M pair | 0.0015 | 1.6 | new | 2.8532 | +0.0865 |
| Baseline (ii): LR 0.003, WD swept | 0.003 | 0.05 | new | 2.7726 | +0.0059 |
| | 0.003 | 0.1 | supplied | 2.7683 | +0.0016 |
| | 0.003 | 0.2 | new | 2.7723 | +0.0056 |

The rule beats reusing the small-scale pair by 0.0865 and ties the best WD in the sweep (+0.0016, within seed noise). Carrying (LR 0.0015, WD 1.6) over without retuning loses because the optimal WD has fallen roughly ×0.5 per doubling; the same LR with WD 0.1 reaches 2.7760, so the WD alone costs 0.077. The test cannot separate the rule from simply keeping WD 0.1, since the loss at LR 0.003 changes by only 0.006 across WD 0.05–0.2.

**Synthesis**

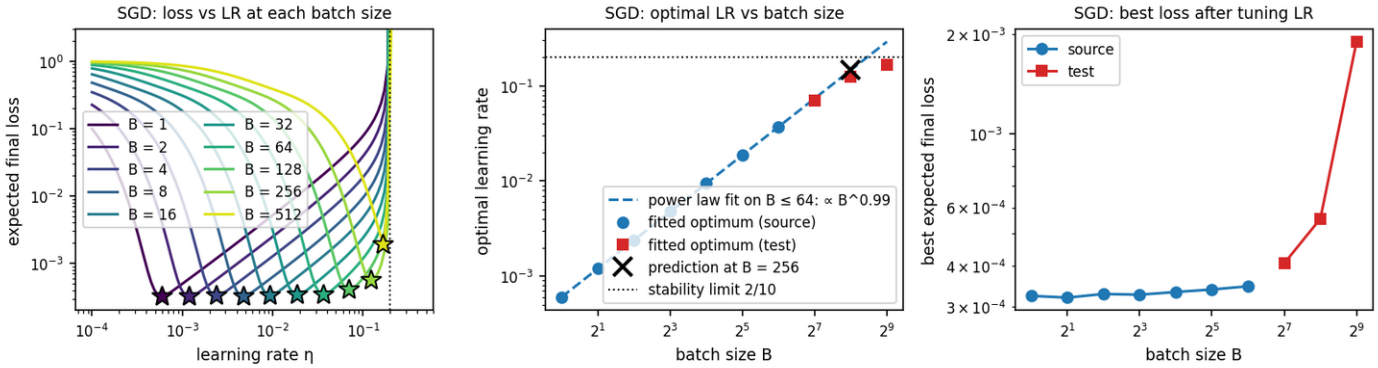
LR and WD act as a coupled pair, and the optimal WD moves with scale much more cleanly than the optimal LR. From 153.6M to 1.2288B tokens the optimal WD falls steadily (∝ D^−0.82, R² 0.994), while the optimal LR rises erratically (R² 0.678) and sits 22–46% below Problem 1's. Raising one partly compensates for lowering the other: moving along $\eta\lambda$ = const is cheaper than changing LR alone at every budget, though the fitted valley is steeper than that line at the three smaller budgets, so the product is a useful summary and not an exact invariant. The benefit of tuning WD shrinks from 0.0447 to 0.0017 as the optimal WD falls toward Problem 1's 0.1. The product law (R² 0.967) was tested at 2.4576B with the LR fixed at 0.003: the predicted WD of 0.0927 beat reusing the 153.6M optimum by 0.087 and tied the best WD in a sweep. Because the loss is nearly flat in WD near the optimum, this shows that WD must be rescaled with budget, not that the product rule beats keeping WD at 0.1. All fits use four budgets, and the rule was checked only one doubling past the data.

## Problem 3: How does batch size affect the hyperparameters?

### 3.1 Noisy quadratic model

Setup: H = diag(1, 10), gradient noise N(0, I/B), N = 8,192 examples, w0 ∼ N(0, diag(1, 0.1)) with 0.1 read as a variance. Each expected final loss averages 4,096 runs; all LRs at a batch size share initializations and noise. LR grid: 100 log-spaced values from 1e−4 to 0.4. Each optimum is a parabola in log η through the lowest grid point and its neighbours.

**(a) SGD**



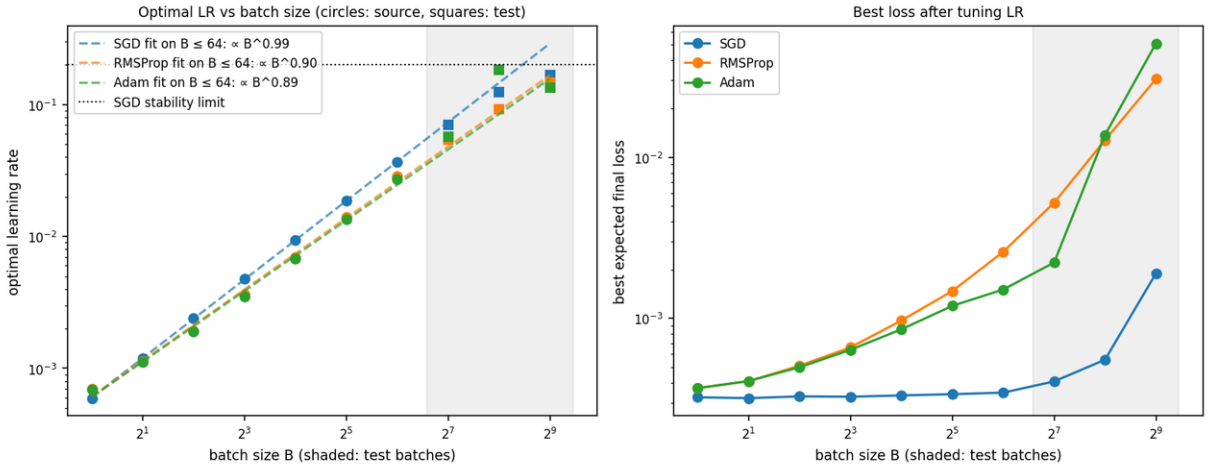

| B | Updates N/B | Fitted optimal η | Best loss |
|---|---|---|---|
| 1 | 8192 | 0.00059 | 0.00032 |
| 2 | 4096 | 0.00120 | 0.00032 |
| 4 | 2048 | 0.00239 | 0.00033 |
| 8 | 1024 | 0.00477 | 0.00033 |
| 16 | 512 | 0.00942 | 0.00033 |
| 32 | 256 | 0.01869 | 0.00034 |
| 64 | 128 | 0.03661 | 0.00035 |
| 128 | 64 | 0.07036 | 0.00041 |
| 256 | 32 | 0.12396 | 0.00055 |
| 512 | 16 | 0.16672 | 0.00190 |

Power law on B ≤ 64: η*(B) ∝ B^0.991. Prediction at B = 256: 0.1463; tuned: 0.1240. Loss at the predicted LR 0.00072, tuned 0.00055.

- **Rule:** the optimal LR doubles with each doubling of the batch, essentially linear scaling.
- **Test at B = 256:** the prediction is 18% above the tuned LR and slightly overshoots.
- **Range:** accurate for B ≤ 64, close at 128 (4% below the line), 15% below at 256. At 512 the line gives 0.29, above the stability limit 2/10, and the optimum is 0.167 (43% below).
- **Noise vs updates:** a larger batch cuts noise per update but also cuts the number of updates. Doubling η with B keeps η × (N/B) at about 4.7–4.8 and the noise floor (∝ η/B) fixed, so the best loss stays flat (about 0.0003) for B ≤ 64. This fails when η hits the stability cap: at B = 512 only 16 updates remain, η × (N/B) falls to 2.7, and most of the 0.0019 loss is leftover error in w1.

**(b) RMSProp and Adam**

RMSProp is Adam with β1 = 0; β2 = 0.95, ε = 10^-8, tuned separately. LR grid: 120 values from 1e−5 to 1.




| B | SGD η* | RMSProp η* | Adam η* | SGD loss | RMSProp loss | Adam loss |
|---|---|---|---|---|---|---|
| 1 | 0.00059 | 0.00070 | 0.00069 | 0.00032 | 0.00037 | 0.00037 |
| 2 | 0.00120 | 0.00113 | 0.00111 | 0.00032 | 0.00041 | 0.00041 |
| 4 | 0.00239 | 0.00194 | 0.00191 | 0.00033 | 0.00051 | 0.00050 |
| 8 | 0.00477 | 0.00362 | 0.00352 | 0.00033 | 0.00066 | 0.00064 |
| 16 | 0.00942 | 0.00693 | 0.00675 | 0.00033 | 0.00097 | 0.00086 |
| 32 | 0.01869 | 0.01410 | 0.01343 | 0.00034 | 0.00148 | 0.00120 |
| 64 | 0.03661 | 0.02864 | 0.02738 | 0.00035 | 0.00259 | 0.00151 |
| 128 | 0.07036 | 0.05451 | 0.05731 | 0.00041 | 0.00522 | 0.00222 |
| 256 | 0.12396 | 0.09272 | 0.18398 | 0.00055 | 0.01277 | 0.01375 |
| 512 | 0.16672 | 0.14659 | 0.13425 | 0.00190 | 0.03057 | 0.05076 |

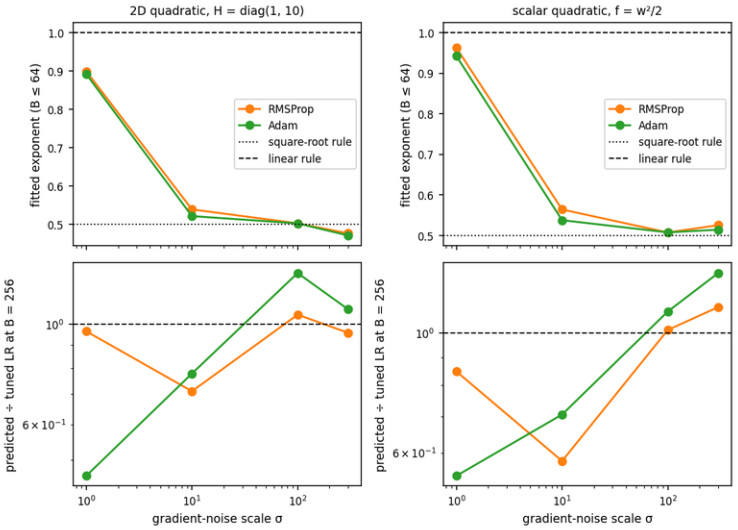
Power laws on B ≤ 64, tested at B = 256:

| Optimizer | Exponent | Predicted LR | Tuned LR | Predicted ÷ tuned | Loss at predicted | Loss at tuned |
|---|---|---|---|---|---|---|
| SGD | 0.991 | 0.1463 | 0.1240 | 1.18 | 0.00072 | 0.00055 |
| RMSProp | 0.899 | 0.0897 | 0.0927 | 0.97 | 0.01278 | 0.01277 |
| Adam | 0.891 | 0.0847 | 0.1840 | 0.46 | 0.01509 | 0.01375 |

- **Exponents:** RMSProp (0.90) and Adam (0.89) match each other and sit slightly below SGD's 0.99; their optima agree within 5% for B ≤ 64 and are about 25% below SGD's for B ≥ 8. They follow a similar rule to each other, slightly sublinear, and a different one from SGD.
- **Transfer at B = 256:** RMSProp's prediction is 3% off with the same loss. Adam's is less than half the tuned LR, but its loss is only 10% worse: its loss is nearly flat for LRs 0.08–0.26 (0.014–0.015), while RMSProp has a sharp minimum near 0.1.
- **Same exponent, different transfer:** Adam's optima drift off the line past the source range (0.057, 0.184, 0.134 at B = 128, 256, 512). The exponent describes the source fit, not the reliability of the extrapolation.
- **Prediction for the language model:** the optimal LR should rise a little slower than linearly (about B^0.9 for Adam-type optimizers) while many updates remain, and degrade at B = 128–256 in the toy (64–32 updates). The language model has 9,375 updates at B = 64, so the rule should hold well above 64, imprecisely. The adaptive optimizers' best loss also worsens with B at fixed examples, unlike SGD's.

**(c) Gradient-noise scale and curvature**

Noise $\sigma^2 I/B$ with $\sigma \in \{1, 10, 100, 300\}$, for RMSProp and Adam on the 2D quadratic and on the scalar quadratic $f(w) = w^2/2$ ($w_0 \sim N(0, 2)$, noise variance $\sigma^2/B$). The LR grid is 150 values from $1e{-}5\sigma$ to $10\sigma$; fit on B ≤ 64, test at B = 256. The σ = 1, 2D rows repeat (b) on a wider grid.

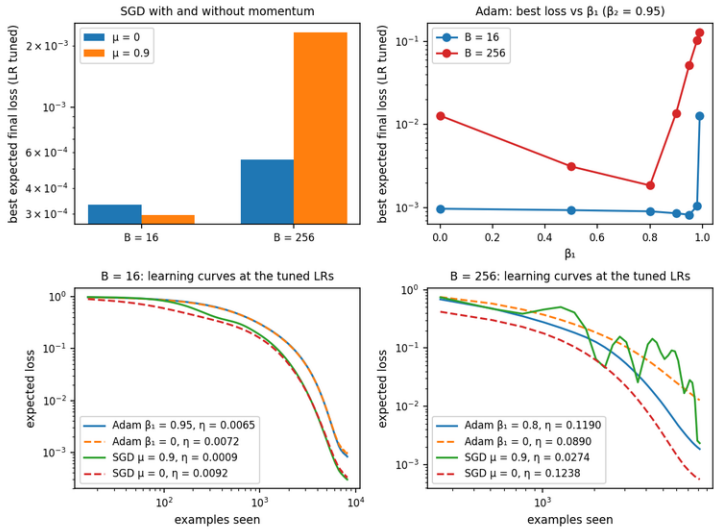


| Model | σ | RMSProp exponent | Adam exponent | RMSProp: LR ratio (predicted / tuned loss) | Adam: LR ratio (predicted / tuned loss) |
|---|---|---|---|---|---|
| 2D | 1 | 0.899 | 0.892 | 0.97 (0.01276 / 0.01283) | 0.46 (0.01508 / 0.01369) |
| 2D | 10 | 0.539 | 0.521 | 0.71 (0.04822 / 0.03717) | 0.78 (0.03918 / 0.03188) |
| 2D | 100 | 0.502 | 0.502 | 1.05 (0.42311 / 0.42300) | 1.30 (0.43507 / 0.43077) |
| 2D | 300 | 0.476 | 0.471 | 0.96 (0.73780 / 0.73782) | 1.08 (0.75978 / 0.75887) |
| Scalar | 1 | 0.962 | 0.942 | 0.85 (0.00432 / 0.00360) | 0.54 (0.01717 / 0.01333) |
| Scalar | 10 | 0.564 | 0.538 | 0.58 (0.05762 / 0.02579) | 0.71 (0.04748 / 0.02978) |
| Scalar | 100 | 0.507 | 0.508 | 1.01 (0.40154 / 0.40153) | 1.10 (0.41132 / 0.40869) |
| Scalar | 300 | 0.525 | 0.514 | 1.12 (0.84012 / 0.83842) | 1.29 (0.87204 / 0.86256) |

- **Signal-to-noise changes the rule.** The exponent falls from about 0.9 at $\sigma = 1$ to 0.5 for $\sigma \geq 10$. Adam divides updates by the root of its second moment, $E[g^2] = (Hw)^2 + \sigma^2/B$, a B-independent signal term plus a noise term falling as 1/B. When the signal dominates the normalizer barely changes with B, and the rule stays close to SGD's (nearly linear). When noise dominates the normalizer is about $\sigma/\sqrt{B}$, so the update acts like SGD with LR $\eta\sqrt{B}/\sigma$; SGD's optimum grows like B, so the Adam-type optimum grows like $\sqrt{B}$.
- **Transfer:** in the noise-dominated regime ($\sigma \geq 100$) the $B = 256$ prediction is within 1.0–1.3× of the tuned LR in both models, with loss within about 1% of tuned. At $\sigma = 1$ and 10 it is less accurate (0.46–0.78 for Adam, 0.58–0.97 for RMSProp) and costs more loss.
- **Non-monotone optimum:** at $\sigma = 300$ the optimal LRs are below those at $\sigma = 100$. The best loss (0.74) is barely under the starting loss (1.0), so little progress is possible in 8,192 examples and a small total step is best.
- **Curvature:** the exponent pattern is the same on the scalar model (0.96/0.94, 0.56/0.54, then 0.51–0.53), so the signal-to-noise conclusion does not depend on curvature. Absolute LRs and small-σ transfer do: scalar optima are larger (0.178 vs 0.061 for RMSProp at $\sigma = 100$, $B = 256$), plausibly because the curvature-10 direction limits the 2D step, and RMSProp transfers worse at $\sigma = 1$ and 10 (0.85 and 0.58 of tuned, vs 0.97 and 0.71), costing more loss at $\sigma = 10$ (0.058 vs 0.026 tuned, against 0.048 vs 0.037).

**(d) Momentum**

$H = \text{diag}(1, 10)$, noise I/B. Entries are the best expected final loss over the LR grid; Adam has $\beta 2 = 0.95$ with β1 and LR swept jointly.




| Optimizer | B = 16: tuned LR, best loss | B = 256: tuned LR, best loss |
|---|---|---|
| SGD, μ = 0 | 0.00922, 0.00033 | 0.1238, 0.00055 |
| SGD, μ = 0.9 | 0.00088, 0.00030 | 0.0274, 0.00234 |

Adam, best loss after tuning the LR at each β1:

| β1 | 0 | 0.5 | 0.8 | 0.9 | 0.95 | 0.98 | 0.99 |
|---|---|---|---|---|---|---|---|
| B = 16 | 0.00097 | 0.00093 | 0.00090 | 0.00086 | **0.00082** | 0.00105 | 0.01267 |
| B = 256 | 0.01277 | 0.00313 | **0.00185** | 0.01375 | 0.05184 | 0.10314 | 0.12911 |

- **SGD:** at B = 16 momentum barely matters once LR is retuned (0.00030 vs 0.00033, about 10%, near the Monte Carlo noise); the tuned LR falls about 10×, so momentum mostly rescales the effective LR $\eta/(1-\mu)$. At B = 256 it hurts (0.00234 vs 0.00055): only 32 updates remain, the buffer needs about 10 to fill, and the curve oscillates.
- **Adam:** the benefit grows strongly with batch size: 18% at B = 16 (best β1 0.95) and 6.9× at B = 256 (best β1 0.8). β1 near one is harmful at the large batch (0.0518 at 0.95), again because only 32 updates remain.
- **Learning curves:** at B = 16 the best-β1 and $\beta 1 = 0$ curves nearly coincide. At B = 256 Adam with $\beta 1 = 0.8$ is below $\beta 1 = 0$ from early on, while SGD with $\mu = 0.9$ oscillates and ends above the no-momentum SGD curve, the lowest of the four.
- **Language model:** the Adam result should carry over, and does in 3.2(c). The SGD result cannot be tested (the language model uses Adam), and the toy's 6.9× is not comparable with language-model loss gaps.

**Synthesis**

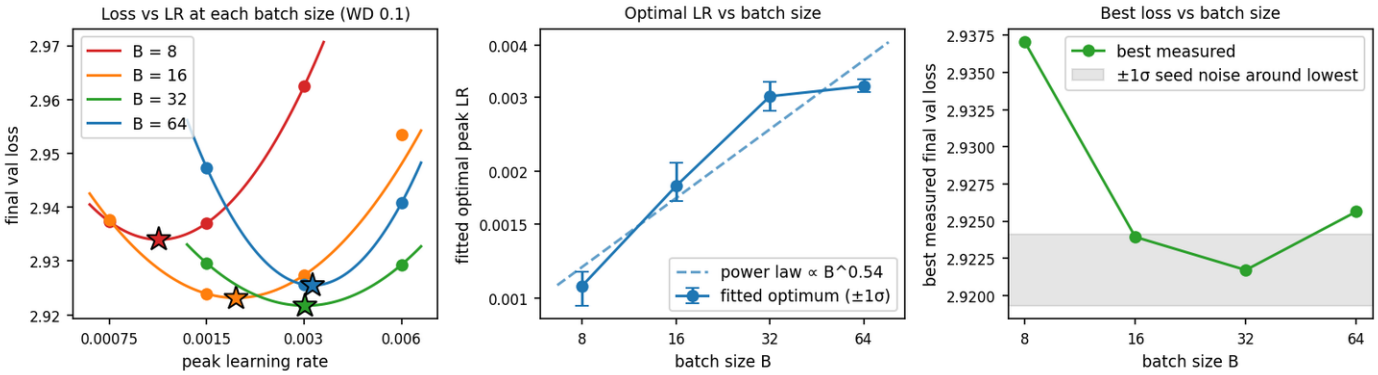
In the noisy quadratic model the optimal LR rises with batch size as a power law whose exponent depends on the optimizer and the signal-to-noise ratio. SGD follows the linear rule (B^0.99) with a flat best loss up to B = 64, then fails as the LR reaches the stability cap (2/10) and too few updates remain (B = 512). RMSProp and Adam follow about B^0.9 when the signal dominates their second moment ($\sigma = 1$) and fall to the square-root rule (about B^0.5) when noise dominates ($\sigma \geq 10$), on both a 2D and a scalar quadratic. A fitted exponent does not guarantee that extrapolation works:

RMSProp's prediction at B = 256 was within 3% of tuned and Adam's 54% off with the same exponent, because Adam's loss was flat there. For momentum, SGD only rescales the effective LR at small batch and hurts at B = 256 with 32 updates, while Adam's benefit grows with batch size (loss ratio 1.18 at B = 16 to 6.9 at B = 256) with a lower best β1. The toy's rules depend on optimizer, noise level and number of updates, and must be checked near the target batch.

### 3.2 Batch size in the language model

All runs: d8, 614.4M tokens (600,000 sequences). B = 8, 16, 32 are new; B = 64 is the supplied P1a / P2a sweeps. Actual tokens processed: 614.4M for B ≤ 64, 614.33M at B = 128 (4,687 updates) and 614.20M at B = 256 (2,343 updates), since incomplete accumulation groups are skipped.

**(a) LR sweeps at WD 0.1**




Final validation loss (lowest per batch size in bold):

| Peak LR | B = 8 | B = 16 | B = 32 | B = 64 |
|---|---|---|---|---|
| 0.00075 | 2.9373 | 2.9377 | | |
| 0.0015 | **2.9371** | **2.9239** | 2.9296 | 2.9473 |
| 0.003 | 2.9624 | 2.9274 | **2.9217** | **2.9256** |
| 0.006 | | 2.9535 | 2.9293 | 2.9408 |

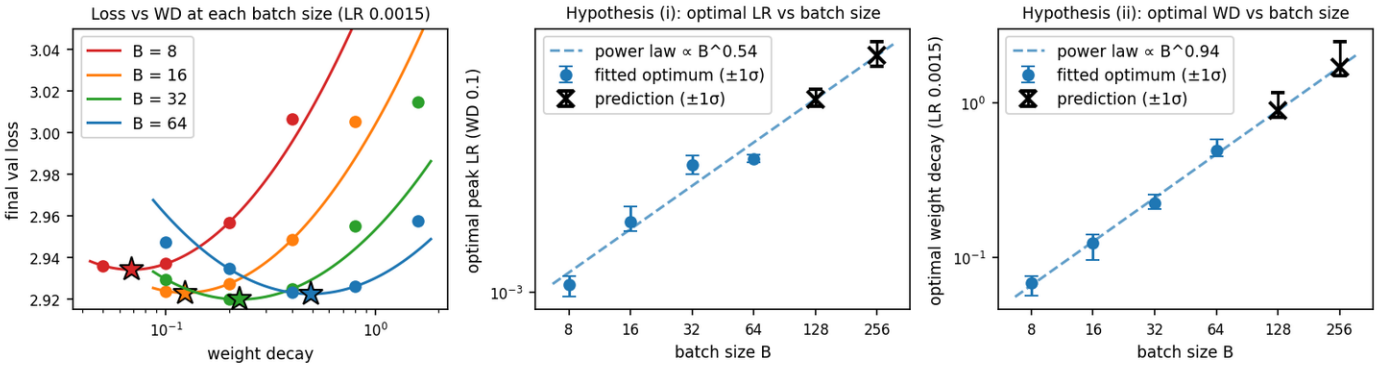
| B | Updates | Fitted optimal LR (±1σ) | Best loss |
|---|---|---|---|
| 8 | 75,000 | 0.00107 (0.00096–0.00115) | 2.9371 |
| 16 | 37,500 | 0.00185 (0.00170–0.00210) | 2.9239 |
| 32 | 18,750 | 0.00302 (0.00278–0.00327) | 2.9217 |
| 64 | 9,375 | 0.00319 (0.00309–0.00331) | 2.9256 |

Power law over the four optima: $\eta^* \propto B^{0.54}$.

- **Optimal LR:** it rises with batch size, steeply from 8 to 32 (about ×1.7 per doubling) and almost flat from 32 to 64 (×1.06). B^0.54 is well below linear, which would put B = 64 near 0.0086; at B = 64, LR 0.006 already costs 0.015 over 0.003.
- **Performance:** it improves from B = 8 to 16–32 and is lowest at B = 32 (2.9217). B = 8 is clearly worse (2.9371, 0.015 above B = 32, many times the seed SD of 0.0024). At B = 64 the loss rises slightly (+0.0039 over B = 32), only about 1σ of the difference between two runs (0.0034), so the loss has flattened rather than clearly turned up. A plausible reading is that the update count has fallen to 9,375 from 75,000, but these runs cannot confirm it.

**(b) Two scaling hypotheses**

(i) fixes WD at 0.1 and scales the optimal peak LR (sources: the LR sweeps above). (ii) fixes LR at 0.0015 and scales the optimal WD (sources: WD sweeps at B = 8, 16, 32 and B = 64). Each is extrapolated to B = 128 and 256 with a power law on B = 8–64.




Final validation loss at LR 0.0015 (lowest per batch size in bold):

| WD | B = 8 | B = 16 | B = 32 | B = 64 |
|---|---|---|---|---|
| 0.05 | **2.9361** | | | |
| 0.1 | 2.9371 | **2.9239** | 2.9296 | 2.9473 |
| 0.2 | 2.9569 | 2.9274 | **2.9201** | 2.9347 |
| 0.4 | 3.0068 | 2.9487 | 2.9250 | **2.9232** |

| WD | B = 8 | B = 16 | B = 32 | B = 64 |
|---|---|---|---|---|
| 0.8 | 3.1031 | 3.0053 | 2.9551 | 2.9261 |
| 1.6 | | 3.1023 | 3.0149 | 2.9576 |

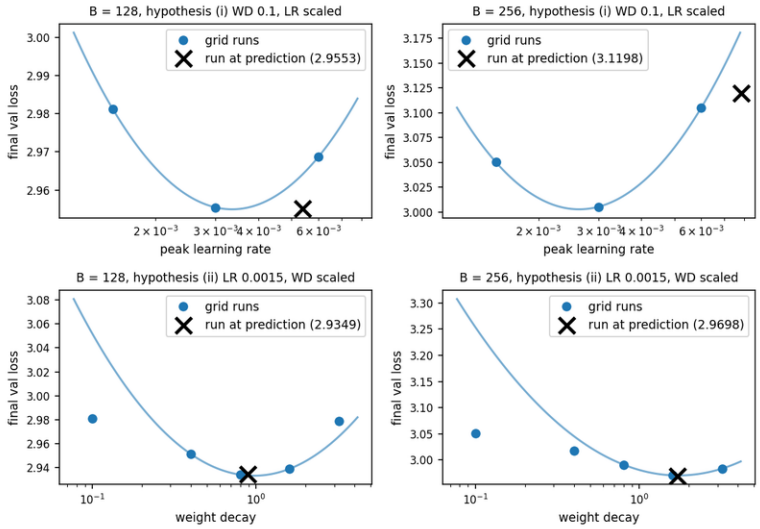
| B | Fitted optimal LR at WD 0.1 (±1σ) | Fitted optimal WD at LR 0.0015 (±1σ) |
|---|---|---|
| 8 | 0.00107 (0.00096–0.00115) | 0.068 (0.057–0.076) |
| 16 | 0.00185 (0.00170–0.00210) | 0.124 (0.096–0.141) |
| 32 | 0.00302 (0.00278–0.00327) | 0.224 (0.206–0.254) |
| 64 | 0.00319 (0.00309–0.00331) | 0.490 (0.447–0.578) |
| Power law | ∝ B^0.54 | ∝ B^0.94 |

At B = 8 and 16 the best measured WD is the lowest tried (0.05, 0.1), so those two WD optima rest on the parabola and could be lower.

**Predictions, recorded before inspecting any B = 128 or 256 run** (±1σ):

| B | (i) Optimal LR at WD 0.1 | (ii) Optimal WD at LR 0.0015 |
|---|---|---|
| 128 | 0.00537 (0.00503–0.00585) | 0.887 (0.80–1.16) |
| 256 | 0.00783 (0.00712–0.00884) | 1.70 (1.49–2.48) |

I trained exactly the predicted configurations and compared them with grid runs (LR {0.0015, 0.003, 0.006} at WD 0.1; WD {0.1, 0.4, 0.8, 1.6, 3.2} at LR 0.0015). Gap = loss at the prediction minus the best sampled loss.

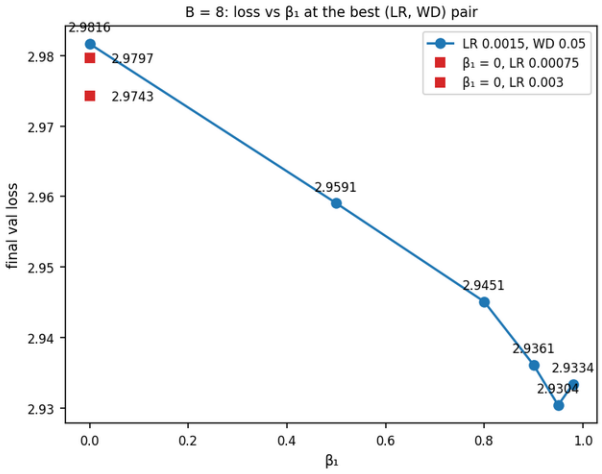


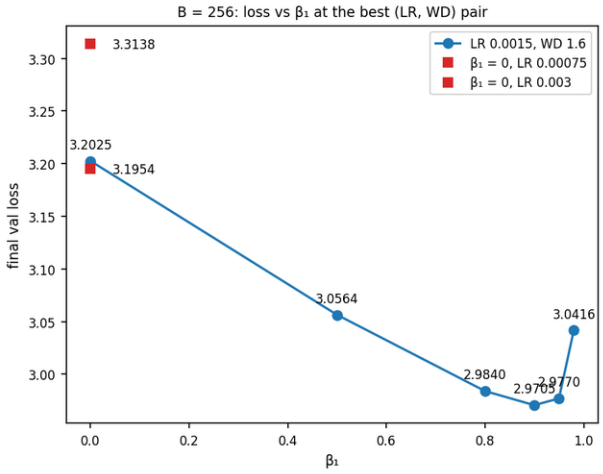

| B | Hypothesis | Predicted config | Fitted optimum | Loss at prediction | Best grid loss in family | Gap in family | Gap to best sampled at this B |
|---|---|---|---|---|---|---|---|
| 128 | (i) LR scaled | LR 0.00537, WD 0.1 | 0.00335 | 2.9553 | 2.9554 (LR 0.003) | −0.0002 | +0.0211 |
| 128 | (ii) WD scaled | LR 0.0015, WD 0.887 | 0.969 | 2.9349 | 2.9342 (WD 0.8) | +0.0007 | +0.0007 |
| 256 | (i) LR scaled | LR 0.00783, WD 0.1 | 0.00263 | 3.1198 | 3.0054 (LR 0.003) | +0.1144 | +0.1493 |
| 256 | (ii) WD scaled | LR 0.0015, WD 1.7 | 1.729 | 2.9698 | 2.9705 (WD 1.6) | −0.0007 | −0.0007 |

- **Hypothesis (ii) transfers better.** Its predicted WD matched the best sampled run at both batch sizes (+0.0007, −0.0007, within seed noise of 0.0024), with fitted optima within 9% and 2% of the predictions. Hypothesis (i) fails at B = 256: the optimal LR at WD 0.1 stopped rising (0.00302, 0.00319, 0.00335, 0.00263 at B = 32 to 256), so the predicted LR was about 3× too high and cost +0.114. Even its best LR at WD 0.1 is 0.021 (B = 128) and 0.036 (B = 256) behind the WD-scaled run, so fixing WD at 0.1 is wrong at large batch whatever the LR.
- **The best configuration:** the LR × WD product at the optimum with LR 0.0015 doubles with each doubling of B (7.4e−4, 1.5e−3, 2.6e−3 at B = 64, 128, 256), consistent with Problem 2: the product is what matters, and WD absorbs the batch-size change.
- **Agreement with the NQM:** partly. The NQM has no weight decay, so it cannot predict that scaling WD works. For the LR, the toy's adaptive optimizers give a sublinear rule that degrades at large batch, and the language model shows a similar shape (B^0.54, then a plateau). The breakdown comes much earlier in update count than in the toy (2,343 updates at B = 256, against 32–64), so fewer updates alone does not explain it.

**(c) Momentum**

LR and WD held at each batch's best measured pair (B = 8: LR 0.0015, WD 0.05; B = 256: LR 0.0015, WD 1.6), $\beta_2 = 0.95$. The $\beta_1 = 0.9$ point is that best run. The $\beta_1 = 0$ control inherited an LR tuned at $\beta_1 = 0.9$, so it is also run at half and double that LR.




| β1 | B = 8 | B = 256 |
|---|---|---|
| 0 | 2.9816 | 3.2025 |
| 0.5 | 2.9591 | 3.0564 |
| 0.8 | 2.9451 | 2.9840 |
| 0.9 | 2.9361 | **2.9705** |
| 0.95 | **2.9304** | 2.9770 |
| 0.98 | 2.9334 | 3.0416 |

β1 = 0 with LR retuned:

| LR | B = 8 | B = 256 |
|---|---|---|
| 0.00075 | 2.9797 | **3.1954** |
| 0.0015 | 2.9816 | 3.2025 |
| 0.003 | **2.9743** | 3.3138 |

- **Effect vs batch size:** momentum is worth about four to five times more at the large batch. β1 = 0 costs 0.051 at B = 8 (2.9816 vs 2.9304 at β1 = 0.95) and 0.232 at B = 256 (3.2025 vs 2.9705 at β1 = 0.9). The best β1 is slightly lower at B = 256 (0.9 vs 0.95), and β1 = 0.98 hurts only at the large batch (3.0416 vs 2.9334).
- **Retuning the control:** the conclusion is unchanged. The best β1 = 0 loss is 2.9743 at B = 8 (LR 0.003) and 3.1954 at B = 256 (LR 0.00075, the lowest tried), so momentum is still worth 0.044 and 0.225. At B = 256 the control's optimum could be a little lower.
- **Versus the toy:** the NQM with Adam also shows a benefit that grows with batch (loss ratio 6.9× at B = 256 vs 1.18× at B = 16) and a lower best β1 at the large batch (0.8 vs 0.95), the same direction as here (0.9 vs 0.95). The sizes differ.

Single seed per setting: differences of a few thousandths at B = 8 (β1 = 0.9, 0.95, 0.98) are within seed noise.

**(d) What the NQM explains**

| Topic | NQM prediction | Language-model measurement |
|---|---|---|
| LR vs batch, SGD | η* ∝ B (B^0.99) while many updates remain | not tested (language model uses Adam) |
| LR vs batch, Adam-type | about B^0.9 if signal dominates, B^0.5 if noise dominates; degrades at large B | η* ∝ B^0.54 for B = 8–64 (0.00107 to 0.00319), then 0.00335 (B = 128) and 0.00263 (B |
| LR prediction at the target batch | B = 256 within 3% (RMSProp) to 54% off (Adam) | predicted 0.00783 vs fitted 0.00263; loss 3.1198 vs 3.0054 at LR 0.003 (+0.114) |
| Weight decay | none | optimal WD at LR 0.0015 ∝ B^0.94; predicted WD gives 2.9349 (B = 128), 2.9698 (B = |
| Momentum | Adam: benefit grows with B (loss ratio 1.18 to 6.9), best β1 falls | β1 = 0 minus best: 0.051 (B = 8), 0.232 (B = 256); best β1 0.95 to 0.9; retuned 0.044 an |
| Best loss at fixed examples | SGD flat then rises; adaptive rises steadily | 2.9371 (B = 8), 2.9239 (16), 2.9217 (32), 2.9256 (64), 2.9342 (128), 2.9705 (256) |

The LR extrapolation beyond B = 64 at fixed WD is a failed prediction: the toy says the rule keeps rising until updates become scarce, but the language model's optimum stops at B = 128–256 with 2,343 updates left. The success of scaling WD is an effect the NQM does not model, since the toy has no weight decay. To model it, add decoupled weight decay to a scale-invariant toy, e.g. f(w) = ½ w^THw / ‖w‖², as normalization layers give: the effective step is η/‖w‖², and weight decay sets ‖w‖ at equilibrium through the product ηλ. That couples LR and WD and says how WD must move with batch size, the effect measured here.

**Synthesis**

In the language model at 614.4M tokens the optimal LR at WD 0.1 rises sublinearly with batch size (B^0.54, from 0.00107 at B = 8 to 0.00319 at B = 64) and then stops (0.00335, 0.00263 at B = 128, 256), so extrapolating the LR rule fails at B = 256: the predicted LR was about three times too high and cost +0.114 in loss. With LR fixed at 0.0015, the optimal WD instead doubles with each doubling of batch (B^0.94), and the WD predicted from B ≤ 64 matched the best sampled configuration at B = 128 and 256 to within 0.0007. The LR × WD product at the optimum rises in proportion to the batch size, tying to Problem 2: the product carries over, not either value alone. Momentum matters more at large batch (β1 = 0 costs 0.044–0.051 at B = 8 and 0.22–0.23 at B = 256, even after retuning LR), with a slightly lower best β1, agreeing with the toy's Adam result. The NQM explains the shape of the Adam-type LR rule and the direction of the momentum effect, but has no weight decay, so it cannot say WD is the better knob to scale, and its breakdown comes much later in update count than in the language model. Best loss at fixed tokens improves from B = 8 to 16–32, is flat to B = 64, and degrades at B = 128 and 256.

## Problem 4: Initialization, feature learning, and transfer

### 4.0 Alignment and scaling rules

Layer $z = n^{-a} W x$, $(W0)_{ij} \sim N(0, \sigma^2 n^{-2b})$, LR $\eta n^{-c}$. Update alignment $R(U, x0) = \Theta(n^{\alpha})$ and readout alignment $S(V0, \Delta x) = \Theta(n^{\omega})$, with $\alpha, \omega \in \{1/2 \text{ (none)}, 1 \text{ (full)}\}$.

- **Hidden (a = 0):** initialization stability gives $a + b = 1/2$, so $b = 1/2$ in all cases. An order-one direct update, $\Theta(\eta n^{(\alpha-a-c)})$, needs $a + c = \alpha$, so $c = \alpha$.
- **Readout (c = 0):** the direct-update condition gives $a = \alpha$. An order-one response to the feature change, $\Theta(n^{(\omega-a-b)})$, needs $a + b = \omega$, so $b = \omega - \alpha$. Initial logits are $\Theta(n^{(1/2-\omega)})$, $O(1)$ for both $\omega$.
- **Input (fixed fan-in d):** the alignment ratio does not grow with n, so $a + b = a + c = 0$.
- **Interaction term:** $O(\eta n^{(\alpha-a-c)}) = O(\eta)$ once $a + c = \alpha$, bounded in all four cases.

| $\alpha$ | $\omega$ | Hidden (a, b, c) | Readout (a, b, c) | Input (a, b, c) |
|---|---|---|---|---|
| 1 (full) | 1 (full) | (0, 1/2, 1) | (1, 0, 0) | (0, 0, 0) |
| 1 (full) | 1/2 (none) | (0, 1/2, 1) | (1, −1/2, 0) | (0, 0, 0) |
| 1/2 (none) | 1 (full) | (0, 1/2, 1/2) | (1/2, 1/2, 0) | (0, 0, 0) |
| 1/2 (none) | 1/2 (none) | (0, 1/2, 1/2) | (1/2, 0, 0) | (0, 0, 0) |

With $m = n/n0$ (readout variance $(1/n0)\, m^{-2b}$, multiplier $m^{-a}$, LR $\eta\, m^{-c}$):

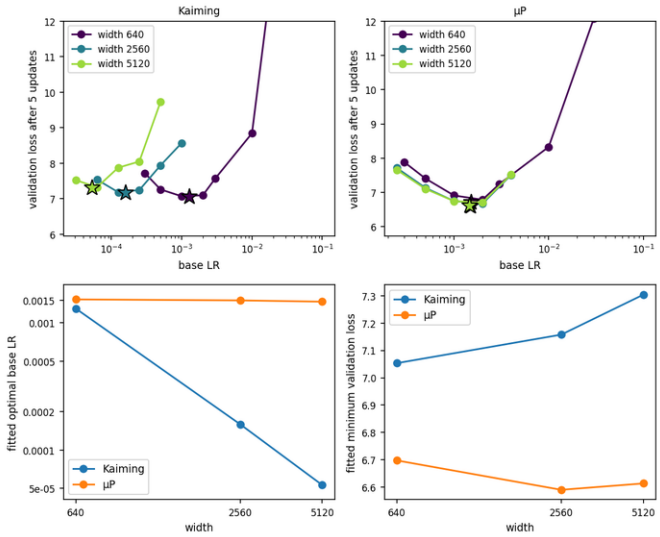
| $\alpha$ | $\omega$ | Hidden: multiplier, variance, LR | Readout: multiplier, variance, LR | Input |
|---|---|---|---|---|
| 1 | 1 | 1, 1/k, η/m | 1/m, 1/n0, η | 1, 1/d, η |
| 1 | 1/2 | 1, 1/k, η/m | 1/m, m/n0, η | 1, 1/d, η |
| 1/2 | 1 | 1, 1/k, $\eta/\sqrt{m}$ | $1/\sqrt{m}$, 1/n, η | 1, 1/d, η |
| 1/2 | 1/2 | 1, 1/k, $\eta/\sqrt{m}$ | $1/\sqrt{m}$, 1/n0, η | 1, 1/d, η |

$\alpha$ sets the hidden LR exponent ($\eta/m$ full, $\eta/\sqrt{m}$ none) and the readout multiplier ($1/m$ or $1/\sqrt{m}$). $\omega$ only sets the readout initialization, through $b = \omega - \alpha$. The hidden initialization and input rules do not depend on either. Full alignment of both recovers the μP rules.

### 4.1 Five-update stress test

**(a) Width transfer**

Prediction (made when choosing the LR grids, after the width-640 pilot): Kaiming's best LR shrinks roughly like 1/width and μP's stays near its width-640 value.



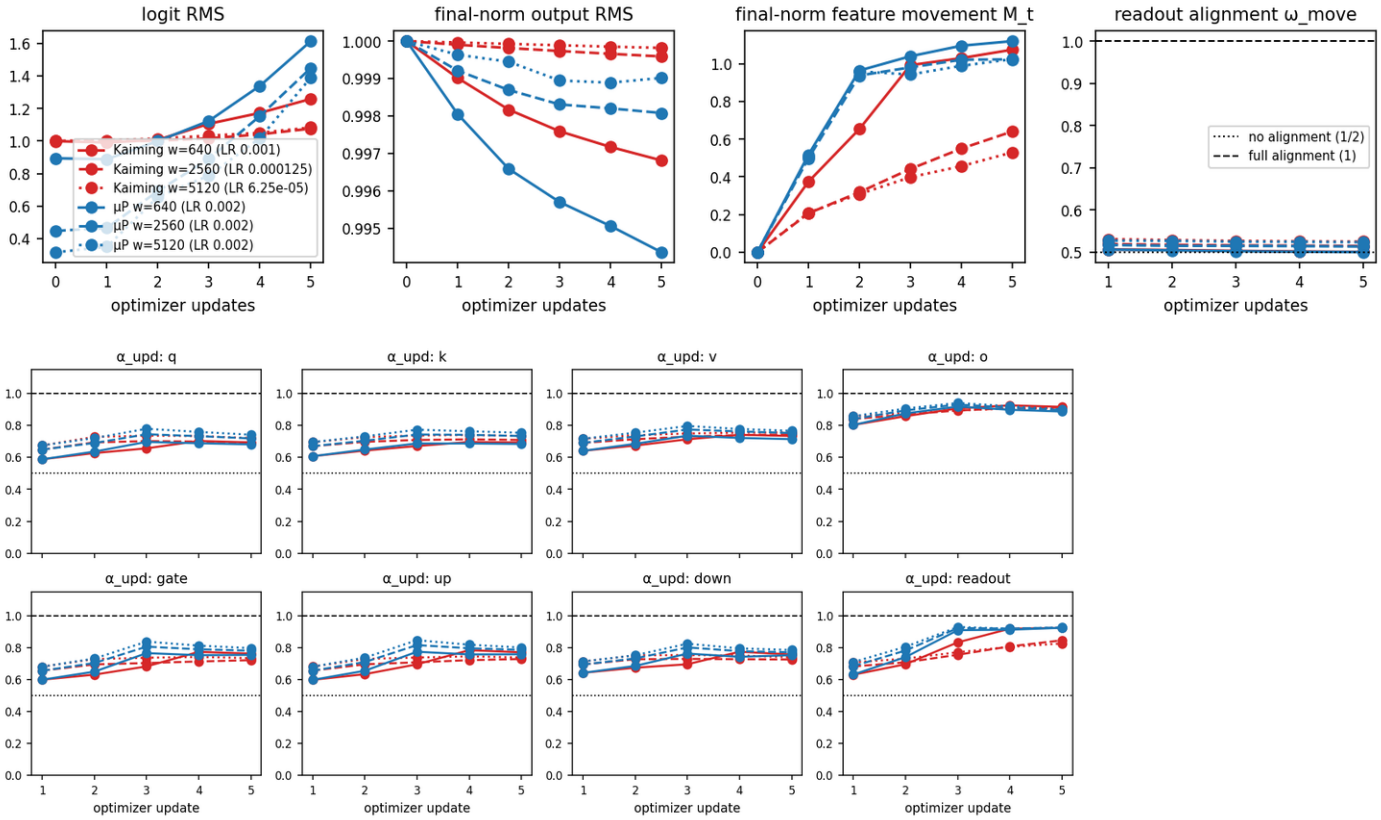

Each curve is a parabola in log LR through the lowest sampled loss and its neighbours. Sweeps are LR: loss.

| Prescription | Width | Fitted η* | Fitted min loss | Sweep |
|---|---|---|---|---|
| Kaiming | 640 | 0.00129 | 7.054 | 0.0003: 7.720, 0.0005: 7.263, 0.001: 7.069, 0.002: 7.098, 0.003: 7.569, 0.01: 8.844, 0.03: 16.324, 0.1: 29.090 |
| Kaiming | 2,560 | 0.00016 | 7.158 | 6.25e−5: 7.539, 1.25e−4: 7.184, 2.5e−4: 7.245, 5e−4: 7.939, 1e−3: 8.571 |
| Kaiming | 5,120 | 0.00005 | 7.304 | 3.1e−5: 7.523, 6.25e−5: 7.325, 1.25e−4: 7.875, 2.5e−4: 8.049, 5e−4: 9.731 |
| μP | 640 | 0.00153 | 6.697 | 0.0003: 7.879, 0.0005: 7.404, 0.001: 6.909, 0.002: 6.784, 0.003: 7.237, 0.01: 8.327, 0.03: 12.085, 0.1: 23.865 |
| μP | 2,560 | 0.00150 | 6.590 | 2.5e−4: 7.710, 5e−4: 7.130, 1e−3: 6.746, 2e−3: 6.669, 4e−3: 7.506 |
| μP | 5,120 | 0.00147 | 6.613 | 2.5e−4: 7.655, 5e−4: 7.101, 1e−3: 6.744, 2e−3: 6.700, 4e−3: 7.518 |

μP transfers its base LR better: the fitted optimum changes by 4% over an 8× increase in width, and the best sampled LR is 0.002 at every width. Kaiming's falls 26× (about width^−1.6, faster than the 1/width predicted). μP also reaches lower loss (6.70, 6.59, 6.61 vs 7.05, 7.16, 7.30), and the gap grows with width (0.36, 0.57, 0.69).

**(b) Width probes**

At each configuration's best sampled LR (Kaiming 0.001, 0.000125, 0.0000625; μP 0.002 at every width).




| Prescription | Width | Logit RMS, step 0 → 5 | M5 | ω_move, step 5 | Mean α_upd, step 5 | Readout α_upd, step 5 |
|---|---|---|---|---|---|---|
| Kaiming | 640 | 1.00 → 1.26 | 1.07 | 0.501 | 0.76 | 0.92 |
| Kaiming | 2,560 | 1.00 → 1.08 | 0.64 | 0.515 | 0.75 | 0.85 |
| Kaiming | 5,120 | 1.00 → 1.08 | 0.53 | 0.525 | 0.76 | 0.83 |
| μP | 640 | 0.89 → 1.61 | 1.12 | 0.500 | 0.75 | 0.93 |
| μP | 2,560 | 0.45 → 1.45 | 1.02 | 0.514 | 0.78 | 0.93 |
| μP | 5,120 | 0.32 → 1.39 | 1.02 | 0.524 | 0.79 | 0.93 |

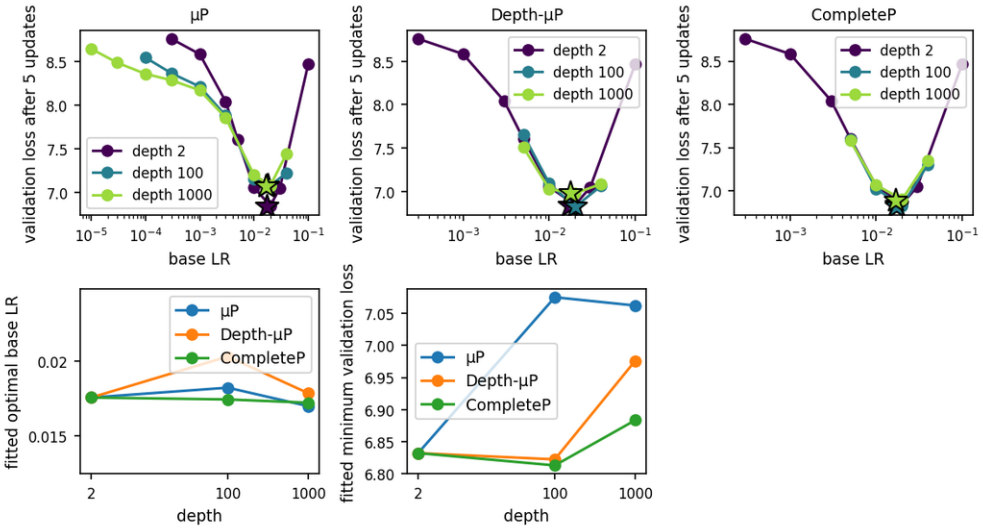
The final RMSNorm output has RMS 1.00 (0.99 at the lowest) everywhere, by construction.

- **Feature movement:** under μP it is width-independent (M1 = 0.52, 0.50, 0.51; M5 = 1.12, 1.02, 1.02), so feature learning is Θ(1). Under Kaiming at its best LR it shrinks with width (M5 = 1.07, 0.64, 0.53), since the LR was cut 8–16×.
- **Logits:** under μP the initial logit RMS falls as m^(−1/2) (0.89, 0.45, 0.32), the Θ(n^(−1/2)) scale derived above, and is order one by update 5 (1.61, 1.45, 1.39). Under Kaiming it starts at 1.0 at every width.
- **Update alignment:** every matrix is between 1/2 and 1, rising over the first three updates then levelling. At update 5, q/k/v are at 0.68–0.77, feedforward at 0.72–0.80, attention output near 0.9, readout 0.83–0.93. Under μP the readout is about 0.93 at every width (near full); under Kaiming it falls with width (0.92, 0.85, 0.83).
- **ω_move:** 0.50–0.53 everywhere, the no-alignment scale, rising slightly with width.
- **Versus the derivation:** the Θ(n^(−1/2)) initial logits, width-independent feature movement under μP, and near-full alignment of the readout update and attention output agree. The assumption that both alignments are full does not: hidden-matrix α_upd is about 0.7–0.8, and ω_move is 1/2. With ω = 1/2 the initial readout does not amplify feature change; the logit change comes from the readout's own update, whose alignment is near full. So μP's flat optimal LR rests on the update alignments, not the initial-readout alignment.

**(c) Depth transfer**

Width 64, one head, reference depth 2, BF16 autocast. At depth 2 (r = 1) all three rules coincide, so one sweep serves all. Standard-μP grids at depths 100 and 1,000 were extended upward after their first sweeps (the loss was still falling at the top of the grid).

Prediction (made when choosing the grids): standard μP needs a smaller LR as depth grows, while Depth-μP and CompleteP keep the depth-2 optimum.

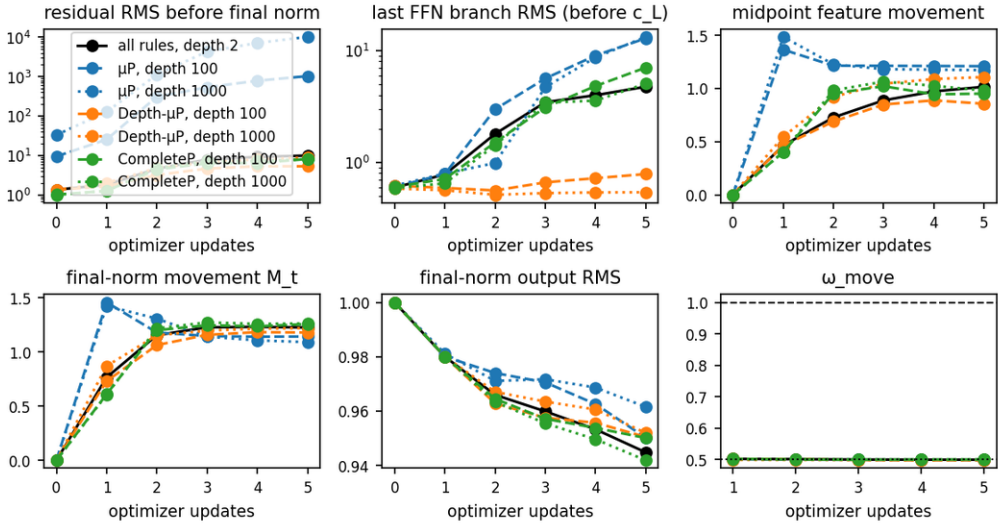



| Prescription | Depth | Fitted η* | Fitted min loss | Sweep (LR: loss) |
|---|---|---|---|---|
| All three | 2 | 0.01738 | 6.832 | 3e−4: 8.759, 1e−3: 8.583, 3e−3: 8.042, 5e−3: 7.607, 0.01: 7.054, 0.02: 6.846, 0.03: 7.048, 0.1: 8.471 |
| Standard μP | 100 | 0.01806 | 7.075 | 1e−4: 8.544, 3e−4: 8.365, 1e−3: 8.211, 3e−3: 7.880, 0.01: 7.157, 0.02: 7.077, 0.04: 7.222 |
| Standard μP | 1,000 | 0.01681 | 7.062 | 1e−5: 8.644, 3e−5: 8.486, 1e−4: 8.357, 3e−4: 8.285, 1e−3: 8.169, 3e−3: 7.855, 0.01: 7.200, 0.02: 7.077, 0.04: 7.445 |
| Depth-μP | 100 | 0.02038 | 6.822 | 0.005: 7.656, 0.01: 7.092, 0.02: 6.823, 0.04: 7.064 |
| Depth-μP | 1,000 | 0.01768 | 6.976 | 0.005: 7.510, 0.01: 7.030, 0.02: 6.979, 0.04: 7.087 |
| CompleteP | 100 | 0.01726 | 6.813 | 0.005: 7.597, 0.01: 7.021, 0.02: 6.828, 0.04: 7.306 |
| CompleteP | 1,000 | 0.01704 | 6.883 | 0.005: 7.580, 0.01: 7.067, 0.02: 6.900, 0.04: 7.354 |

- **LR transfer:** the prediction that standard μP would lose transfer is not supported. All three keep their fitted optimum within 17% of the depth-2 value out to depth 1,000 (ratios 1.04, 0.97 standard; 1.17, 1.02 Depth-μP; 0.99, 0.98 CompleteP), and the best sampled LR is 0.02 throughout. The grid is coarse (factor 2), so differences under about 40% are not resolved.
- **Loss:** better LR transfer does not imply lower loss here, since transfer is about equal and the losses differ. At depth 100 the fitted minimum is 7.075 (standard) vs 6.822 (Depth-μP) and 6.813 (CompleteP); at depth 1,000, 7.062, 6.976 and 6.883. The ranking is CompleteP, Depth-μP, standard μP, and standard μP's loss worsens with depth (6.832 to 7.075).

**(d) Depth probes**

At each prescription's best sampled LR (0.02 everywhere).




| Prescription | Depth | c_L | Residual RMS before the final norm, step 0 → 5 | Branch RMS (last block, before c_L), step 5 | M5 final norm | M5 midpoint | Logit RMS, step 5 |
|---|---|---|---|---|---|---|---|
| All three | 2 | 1 | 1.34 → 9.9 | 4.75 | 1.22 | 1.02 | 1.91 |
| Standard μP | 100 | 1 | 9.4 → 1,030 | 12.9 | 1.14 | 1.21 | 2.07 |
| Standard μP | 1,000 | 1 | 33 → 10,200 | 13.3 | 1.09 | 1.17 | 2.02 |
| Depth-μP | 100 | 0.141 | 1.34 → 5.3 | 0.79 | 1.18 | 0.86 | 1.83 |
| Depth-μP | 1,000 | 0.045 | 1.34 → 8.8 | 0.54 | 1.22 | 1.11 | 2.17 |
| CompleteP | 100 | 0.020 | 1.01 → 8.1 | 7.04 | 1.26 | 0.95 | 1.92 |
| CompleteP | 1,000 | 0.002 | 1.00 → 8.0 | 5.06 | 1.26 | 0.98 | 1.88 |

- **Normalized features can hide a growing residual stream.** Normalized features look the same for all three prescriptions at every depth (RMS 0.94–0.96 after five updates, movement 0.86–1.26, logit RMS 1.8–2.2). Yet under standard μP the residual stream grows from 9.9 at depth 2 to 1,030 at depth 100 and 10,200 at depth 1,000 (and starts at 9.4 and 33), while under Depth-μP and CompleteP it stays at about 5–9. The RMS

normalization before each block hides this growth.

- **ω_move** is 0.50 for every prescription and depth through all five updates.
- **Link to (c):** the normalized signals are bounded and the feature change is order one for all three, consistent with all three transferring the LR. The unbounded residual stream under standard μP costs loss (0.18–0.25 above the depth-aware rules), not LR transfer, in this five-update test, which may be too short for its instability to move the optimum.

**Synthesis**

In the five-update test μP transfers the base LR across width and Kaiming does not: μP's fitted optimum is 0.0015 at widths 640, 2,560 and 5,120 (4% spread) while Kaiming's falls 26× (0.00129 to 0.00005), and μP reaches lower loss (6.61 vs 7.30 at width 5,120). The probes agree with the derivation on what μP is built to fix: initial logits fall as n^(−1/2) and feature movement is width-independent (M5 ≈ 1.0–1.1) under μP but shrinks under Kaiming at its best LR. They do not support the assumption of full alignment on both sides: hidden update alignment is about 0.7–0.8 and ω_move about 0.5, so μP works through the update alignments and the readout's own update. Across depth all three prescriptions kept the optimal LR (0.017–0.020) from depth 2 to 1,000, so LR transfer did not separate them; the loss did (CompleteP < Depth-μP < standard μP at depth 1,000: 6.88, 6.98, 7.06). The normalized features look alike while the standard-μP residual stream grows to about 10^4, so similar normalized activations can hide a growing residual stream, and the cost shows in loss rather than the optimal LR within five updates.

### 4.2 Longer training (153.6M tokens)

Course model, AdamW, WD 0.1, $\beta = (0.9, 0.95)$, linear decay with 1% warmup, clipping at 1 (2,344 updates). μP uses $n0 = 512$: embedding $G/n0$, readout $G/\sqrt{n0}$, readout multiplier $1/m$, hidden LR $\eta/m$. Depth runs: width 512, reference depth 8, $r = L/8$. The width-512 curve is the supplied Problem 1 sweep, the same model under both prescriptions ($m = 1$).

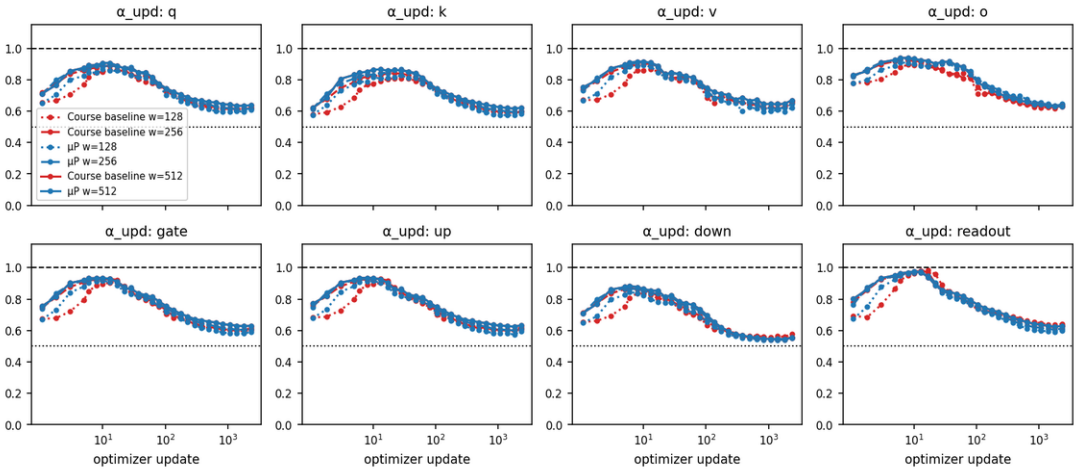
**(a) Transferring the reference LR**

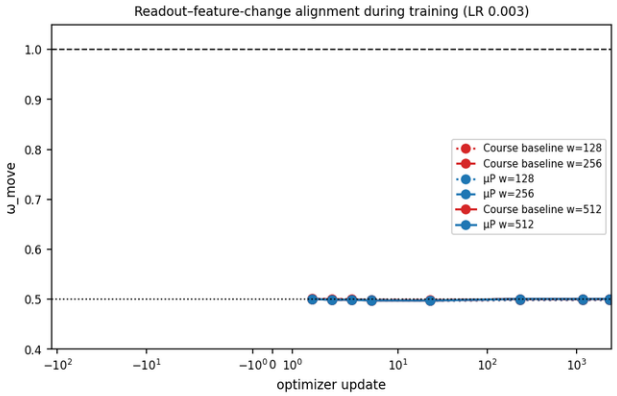
Prediction (reflected in the LR grids chosen before launching): Problem 1's best LR (0.003) transfers under μP but not under the baseline, whose optimum I expected to rise at narrower widths (so its grid extends to 0.024).

Final validation loss (LR: loss):

| Prescription | Width | Sweep |
|---|---|---|
| Baseline | 128 | 0.0015: 3.8826, 0.003: 3.6882, 0.006: **3.6241**, 0.012: 3.6363, 0.024: 3.6484 |
| Baseline | 256 | 0.0015: 3.4891, 0.003: **3.4094**, 0.006: 3.4126, 0.012: 3.4442, 0.024: 3.5200 |
| μP | 128 | 0.00075: 3.7658, 0.0015: 3.6510, 0.003: **3.6293**, 0.006: 3.6476, 0.012: 3.7062 |
| μP | 256 | 0.00075: 3.5125, 0.0015: 3.4126, 0.003: **3.4116**, 0.006: 3.4423, 0.012: 3.4900 |
| Both (supplied) | 512 | 0.0015: 3.2442, 0.003: **3.2291**, 0.006: 3.3056 |

Under μP the source LR transfers at both widths (0.003 is best sampled). Under the baseline it transfers to width 256 but not 128, where 0.006 is better by 0.064.

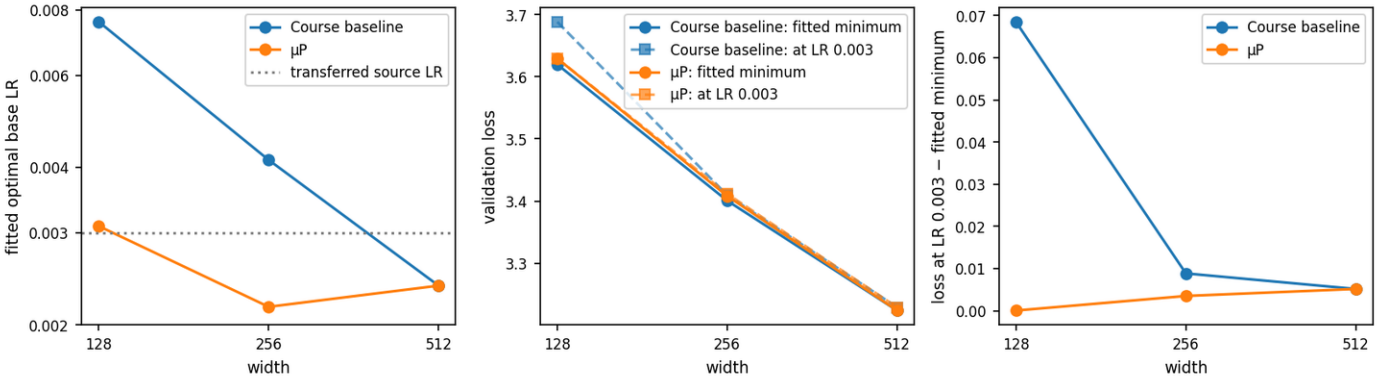


| Prescription | Width | Mean α_upd, update 5 → end | Readout α_upd, update 5 → end | ω_move range |
|---|---|---|---|---|
| Baseline | 128 | 0.77 → 0.62 | 0.86 → 0.64 | 0.497–0.500 |
| Baseline | 256 | 0.87 → 0.61 | 0.95 → 0.63 | 0.499–0.501 |
| μP | 128 | 0.85 → 0.60 | 0.92 → 0.60 | 0.497–0.501 |
| μP | 256 | 0.88 → 0.61 | 0.94 → 0.61 | 0.497–0.501 |
| Both | 512 | 0.89 → 0.63 | 0.95 → 0.63 | 0.497–0.501 |

α_upd keeps rising for the first 10–20 updates to a peak of about 0.85–0.97 (attention output, gate and up near 0.92; q, k, v, down 0.82–0.9; readout highest at 0.97–0.98), then decays to about 0.55–0.65 by the end. The shape is the same for both prescriptions and all widths; the narrowest baseline just rises more slowly. ω_move stays at 0.50 (0.497–0.503) at every checkpoint, width and prescription.

**(b) Transfer vs tuned performance**

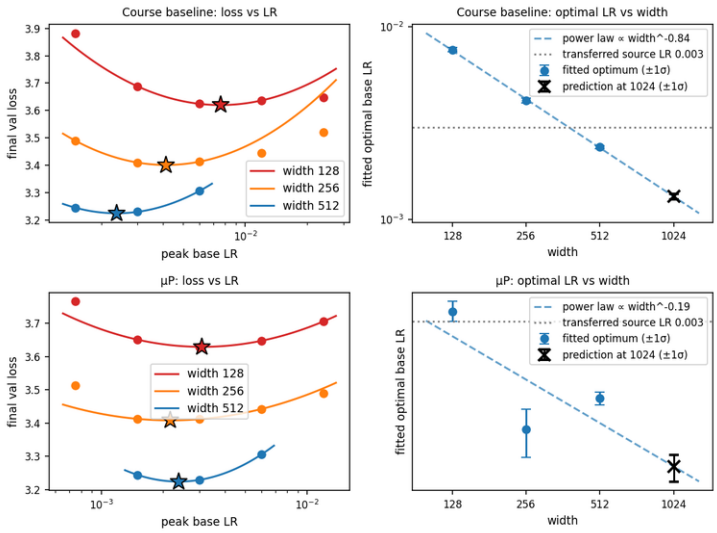



| Prescription | Width | Fitted η* | Fitted min loss | Loss at LR 0.003 | Gap |
|---|---|---|---|---|---|
| Baseline | 128 | 0.00759 | 3.6197 | 3.6882 | +0.0685 |
| Baseline | 256 | 0.00413 | 3.4006 | 3.4094 | +0.0088 |
| Baseline | 512 | 0.00238 | 3.2240 | 3.2291 | +0.0052 |
| μP | 128 | 0.00309 | 3.6293 | 3.6293 | +0.0000 |
| μP | 256 | 0.00217 | 3.4082 | 3.4116 | +0.0035 |
| μP | 512 | 0.00238 | 3.2240 | 3.2291 | +0.0052 |

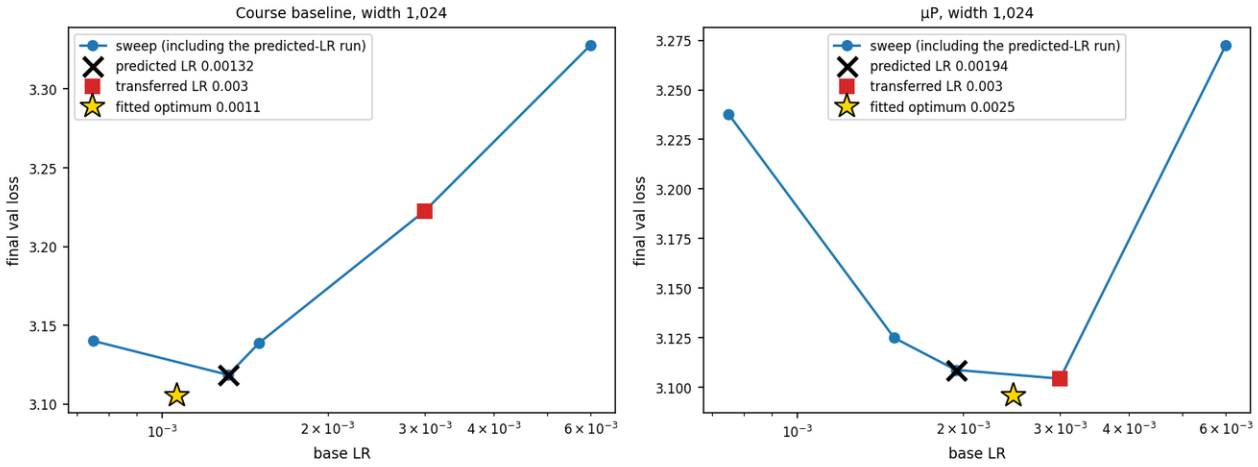
μP transfers better (its optimum stays within 0.00217–0.00309 and the transferred LR costs at most 0.005; the baseline's falls from 0.00759 to 0.00238, about width^−0.84, costing 0.069 at width 128), but not lower after tuning: the baseline is slightly lower at widths 128 and 256 (3.6197 vs 3.6293; 3.4006 vs 3.4082), by 0.008–0.010, a few times the expected pairwise seed noise (about 0.0034, from A1 at 614M tokens). They are identical at 512. After five updates μP was better in both respects (loss lower by 0.36–0.69); over thousands of updates the tuned losses are nearly equal and the transfer benefit is concentrated at the narrowest width (baseline penalty 0.069, 0.009, 0.005 at widths 128, 256, 512).

**(c) Held-out width 1,024**

**Prediction, recorded before inspecting any width-1,024 run:** a power law fitted to the fitted optima at widths 128, 256, 512 (bands ±1σ from seed-noise refits, SD 0.0024).




| Prescription | η* at 128 | η* at 256 | η* at 512 | Power law | Predicted η* at 1,024 |
|---|---|---|---|---|---|
| Baseline | 0.00759 | 0.00413 | 0.00238 | ∝ width^−0.84 | 0.00132 (0.00127–0.00136) |
| μP | 0.00309 | 0.00217 | 0.00238 | ∝ width^−0.19 | 0.00194 (0.00185–0.00200) |



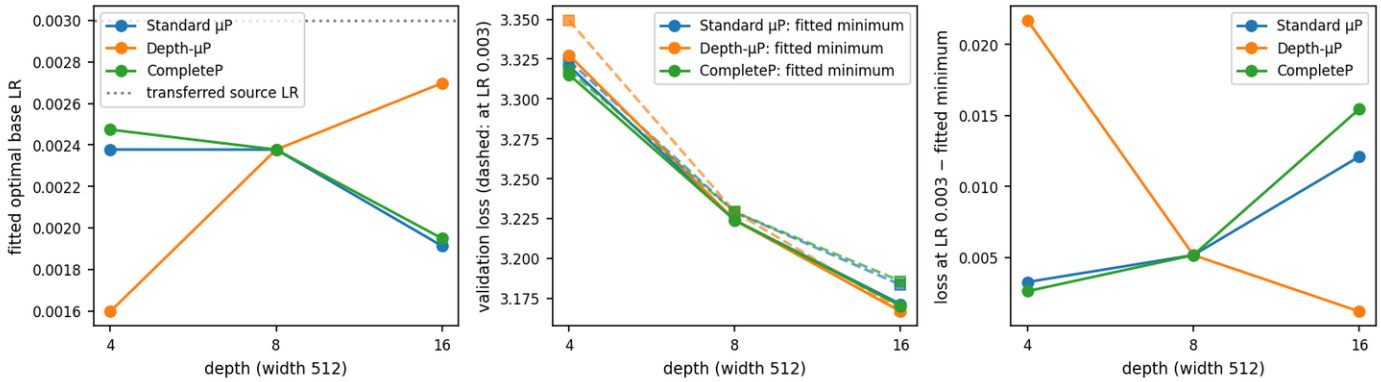

| Prescription | Predicted LR | Fitted local optimum | Best sampled LR (loss) | Loss at predicted (gap) | Loss at transferred 0.003 (gap) |
|---|---|---|---|---|---|
| Baseline | 0.00132 | 0.00106 | 0.00132 (3.1186) | 3.1186 (+0.0000) | 3.2226 (+0.1040) |
| μP | 0.00194 | 0.00247 | 0.003 (3.1044) | 3.1088 (+0.0044) | 3.1044 (+0.0000) |

Sweeps (LR: loss): baseline 0.00075: 3.1402, 0.00132: 3.1186, 0.0015: 3.1388, 0.003: 3.2226, 0.006: 3.3278; μP 0.00075: 3.2378, 0.0015: 3.1249, 0.00194: 3.1088, 0.003: 3.1044, 0.006: 3.2726.

The predictions are 25% above the fitted optimum for the baseline and 21% below it for μP, within the resolution of a factor-2 grid. Fitting the width dependence helps the baseline a lot (3.1186 vs 3.2226 for the transferred LR, which is about 3× too high at width 1,024) and does not help μP, where the transferred LR was the best run and the prediction is 0.0044 behind (within noise), since μP's optimum barely moves. After tuning at width 1,024 μP is 0.014 better (3.1044 vs 3.1186), the reverse of the small baseline advantage at widths 128 and 256.

**(d) Depth transfer**

Width 512, depths 4, 8 (supplied; all rules coincide) and 16, LRs {0.0015, 0.003, 0.006}, parabola through the three points.




| Prescription | Depth | Sweep (LR: loss) | Fitted η* | Fitted min loss | Loss at 0.003 |
|---|---|---|---|---|---|
| All three | 8 | 0.0015: 3.2442, 0.003: 3.2291, 0.006: 3.3056 | 0.00238 | 3.2240 | 3.2291 |
| Standard μP | 4 | 0.0015: 3.3335, 0.003: 3.3240, 0.006: 3.3724 | 0.00238 | 3.3207 | 3.3240 |
| Standard μP | 16 | 0.0015: 3.1750, 0.003: 3.1835, 0.006: 3.2496 | 0.00191 | 3.1714 | 3.1835 |
| Depth-μP | 4 | 0.0015: 3.3278, 0.003: 3.3493, 0.006: 3.4233 | 0.00160 | 3.3275 | 3.3493 |
| Depth-μP | 16 | 0.0015: 3.2029, 0.003: 3.1680, 0.006: 3.2337 | 0.00270 | 3.1668 | 3.1680 |
| CompleteP | 4 | 0.0015: 3.3329, 0.003: 3.3179, 0.006: 3.3704 | 0.00247 | 3.3153 | 3.3179 |
| CompleteP | 16 | 0.0015: 3.1761, 0.003: 3.1859, 0.006: 3.2755 | 0.00195 | 3.1704 | 3.1859 |

A depth correction does not clearly help. From depth 8 to 16 the fitted optimum moves ×0.80 (standard), ×1.13 (Depth-μP), ×0.82 (CompleteP); from 8 to 4, ×1.00, ×0.67, ×1.04. The loss gap from not retuning is +0.012, +0.001, +0.015 at depth 16 and +0.003, +0.022, +0.003 at depth 4, so Depth-μP transfers best at 16 and worst at 4. Tuned minimum losses differ by at most 0.007 at each depth (depth 16: 3.1714, 3.1668, 3.1704), about seed noise. A factor-2 change in depth is too small to separate the prescriptions, consistent with standard μP also keeping its LR in the five-update test.

**(e) Training regime**

Logit RMS, final-norm output movement divided by its RMS (normalized feature movement) and ω_move on fixed inputs at update 5 and at the end, at the transferred and best sampled LRs. The pre-clip gradient norm is logged at nine checkpoints; update 1 is shown.

| Configuration | LR | Final loss | Logit RMS (5 / end) | Norm. feature movement (5 / end) | ω_move (5 / end) | Grad norm at update 1 (clip coef.) |
|---|---|---|---|---|---|---|
| Baseline, width 128 | 0.003 (transferred) | 3.6882 | 0.97 / 4.37 | 1.29 / 1.16 | 0.498 / 0.499 | 3.8 (0.26) |
| Baseline, width 128 | 0.006 (best) | 3.6241 | 0.95 / 4.48 | 1.37 / 1.19 | 0.497 / 0.499 | 3.8 (0.26) |
| Baseline, width 256 | 0.003 (best) | 3.4094 | 0.94 / 4.21 | 1.20 / 1.24 | 0.499 / 0.501 | 8.8 (0.11) |
| Baseline, width 1,024 | 0.003 (transferred) | 3.2226 | 1.00 / 4.18 | 1.33 / 1.91 | 0.500 / 0.501 | 33.8 (0.03) |
| Baseline, width 1,024 | 0.00132 (best) | 3.1186 | 0.96 / 3.97 | 1.33 / 1.44 | 0.499 / 0.502 | 33.8 (0.03) |
| μP, width 128 | 0.003 (best) | 3.6293 | 1.84 / 4.45 | 1.37 / 1.67 | 0.498 / 0.499 | 29.2 (0.03) |

| Configuration | LR | Final loss | Logit RMS (5 / end) | Norm. feature movement (5 / end) | ω_move (5 / end) | Grad norm at update 1 (clip coef.) |
|---|---|---|---|---|---|---|
| μP, width 256 | 0.003 (best) | 3.4116 | 1.28 / 4.29 | 1.25 / 1.51 | 0.499 / 0.501 | 25.3 (0.04) |
| μP, width 1,024 | 0.003 (best) | 3.1044 | 0.73 / 4.09 | 1.32 / 1.39 | 0.501 / 0.503 | 11.8 (0.08) |
| Standard μP, depth 16 | 0.003 (transferred) | 3.1835 | 0.95 / 4.17 | 1.30 / 1.45 | 0.500 / 0.501 | 19.0 (0.05) |
| Standard μP, depth 16 | 0.0015 (best) | 3.1750 | 0.96 / 4.00 | 1.32 / 1.30 | 0.500 / 0.500 | 19.0 (0.05) |
| Depth-μP, depth 4 | 0.003 (transferred) | 3.3493 | 0.95 / 4.29 | 1.35 / 1.45 | 0.500 / 0.500 | 14.7 (0.07) |
| Depth-μP, depth 4 | 0.0015 (best) | 3.3278 | 0.96 / 4.13 | 1.36 / 1.30 | 0.500 / 0.500 | 14.7 (0.07) |
| Depth-μP, depth 16 | 0.003 (best) | 3.1680 | 0.95 / 4.17 | 1.30 / 1.45 | 0.500 / 0.501 | 19.0 (0.05) |
| CompleteP, depth 16 | 0.003 (transferred) | 3.1859 | 0.95 / 4.18 | 1.30 / 1.45 | 0.500 / 0.501 | 19.0 (0.05) |
| CompleteP, depth 16 | 0.0015 (best) | 3.1761 | 0.96 / 3.97 | 1.32 / 1.30 | 0.500 / 0.501 | 19.0 (0.05) |

The supplied width-512 reference (baseline, also μP) starts at 15.8 (coefficient 0.063), with the embedding 98% of the squared gradient norm at the start and 3% by the end.

- **Alignment assumptions:** update–input alignment is close to full early (α_upd 0.85–0.98 within the first 10–20 updates at every width, both prescriptions) and relaxes to about 0.6 by the end, so full alignment describes early updates only. The readout–feature-change alignment is never near full: ω_move is 0.50 at every width, depth, prescription and stage.
- **Larger early transients, better final loss:** yes. At width 128, μP has a much larger first gradient (29.2 vs 3.8; clip coefficient 0.03 vs 0.26) and larger logit change by update 5 (1.84 vs 0.97), yet ends within 0.01 of the baseline's tuned loss (3.6293 vs 3.6241). Conversely, at width 1,024 the baseline at the transferred and best LRs looks the same after five updates (logit RMS 1.00 vs 0.96, movement 1.33 vs 1.33), but the transferred run ends with much larger feature movement (1.91 vs 1.44) and a loss 0.104 worse. End-of-training feature movement separates them, not the early transients.
- **Does an LR shift alone show the model left the stable regime?** No. Standard μP's optimum moves ×0.8 from depth 8 to 16 while its logits, feature movement, α_upd and ω_move match the other depths; the baseline's optimum moves 3× across widths 128–512 while its alignment curves match at every width. The one configuration that stands out in the diagnostics (baseline, width 1,024, transferred LR) is also clearly worse in loss.
- **Normalization and clipping:** the update-1 gradient norm depends on width in opposite directions: for the baseline it grows with width (3.8, 8.8, 15.8, 33.8 at widths 128, 256, 512, 1,024; clip coefficient 0.26 down to 0.03), for μP it falls (29.2, 25.3, 15.8, 11.8). The embedding dominates early, and clipping is active only for roughly the first 20 updates (coefficient 1.0 by update 234). It affects early dynamics differently for the two prescriptions but cannot explain late differences; I ran no clipping or normalization ablations.
- **LR–WD coupling:** WD is fixed at 0.1. Under μP the hidden decay per step, $(\eta/m)\lambda$, is 4× larger at width 128 and 2× smaller at 1,024 than at 512, while for the baseline it is the same at every width. Problem 2 found tuning WD at 153.6M tokens worth 0.045 at width 512 (LR 0.0015, WD 1.6), several times the largest prescription difference after LR tuning (0.010), so a fixed WD could mask or exaggerate the comparison.

**Synthesis**

At 153.6M tokens the μP advantage seen after five updates mostly shrinks. Problem 1's best LR (0.003) transfers under μP from width 128 to 1,024, where the fitted optimum stays between 0.0022 and 0.0031, while the baseline's falls (0.0076, 0.0041, 0.0024 at widths 128, 256, 512) and the transferred LR costs it 0.069 at width 128 and 0.104 at width 1,024. Tuned losses are nearly equal: the baseline is lower by 0.008–0.010 at widths 128 and 256, μP lower by 0.014 at width 1,024. Fitting the baseline's width dependence works for the held-out width (the predicted LR was the best sampled run, 0.104 better than direct transfer) and adds nothing for μP. Depth corrections did not clearly help at depths 4 and 16 (the optimum moved by at most 33%, tuned losses differ by at most 0.007). The diagnostics show why the derivation's alignment assumptions are only partly right: update–input alignment is near full early (0.85–0.98 at updates 10–20) and relaxes to about 0.6, and the readout–feature-change alignment stays at 1/2 at every width, depth and stage. An LR shift alone does not show a model has left the stable regime, and the early gradient norm and clipping depend on width in opposite directions for the two prescriptions. Because WD was fixed and Problem 2 showed tuning it matters more at this budget (0.045) than the prescription differences, a joint LR and WD sweep would be needed for a clean comparison.